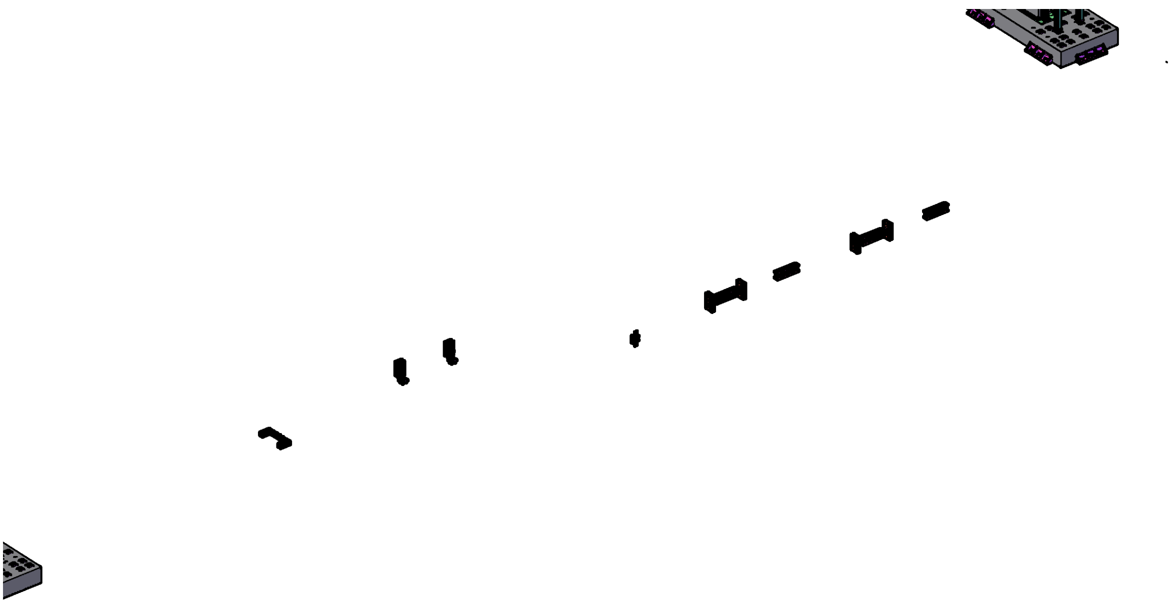
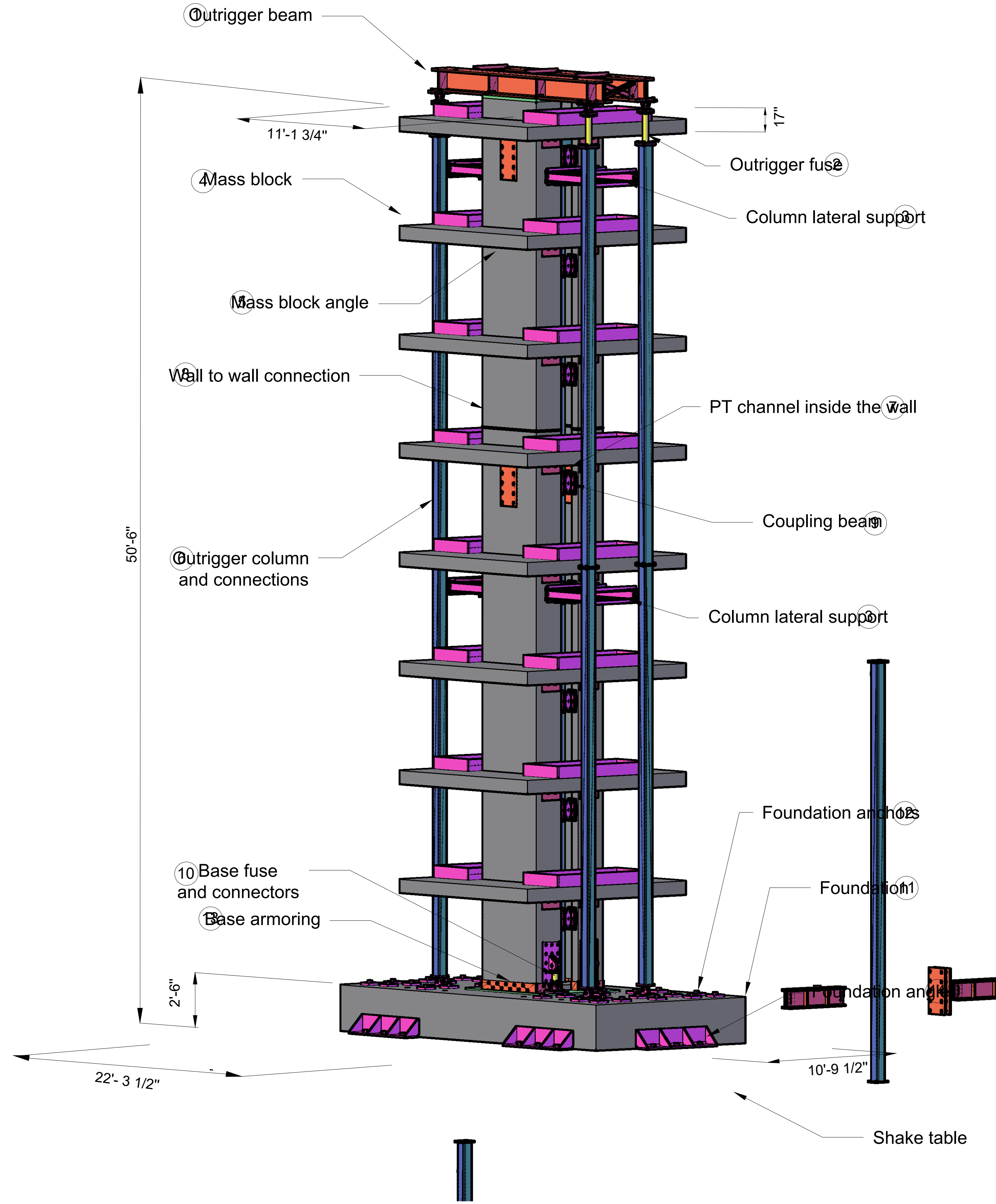


Components

Num	Component	Drawing	Qty	Location of Fabrication
1	Outrigger beam	P13	2	GTS Canada
2	Outrigger fuse and connectors	P16	4	GTS Canada
3	Column lateral support	P15	4	GTS Canada
4	Mass block	P04,P07	8	China
5	Mass block angle	P04,P07	32	GTS Canada
6	Outrigger column and connections	P16,P17	4	GTS Canada
7	PT channels inside the wall	P12	4	GTS Canada
8	Wall to wall connection	P14	2	Canada
9	Coupling beam	P09	32	GTS Canada
10	Base fuse and connectors	P10	4	GTS Canada
11	Foundation	P03	1	China
12	Foundation anchors	P03	12	GTS Canada
13	Base armoring	P11	2	GTS Canada
14	Foundation angle	P03	6	GTS Canada



Material Properties

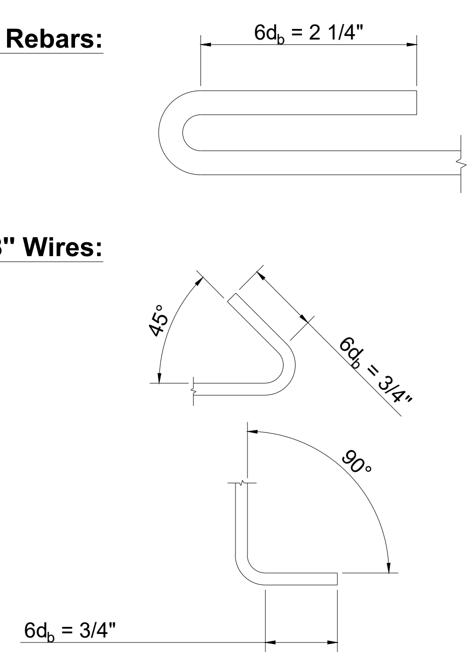
Unless otherwise specified in the drawings, materials for all components should be:

Steel:
ASTM A36 for steel plates;
G40.21 300W for angles;
ASTM A572- Gr50 for hot-rolled W, M, HP, and channel sections
ASTM A500 - Gr50 for HSS sections
Threaded Rods:
ASTM F1554-Gr105 for threaded rods (no welding required);
ASTM F1554-Gr55 for threaded rods (welding required);
Bolts:
ASTM A490 for bolts to connect coupling beams
ASTM A325 for other bolts
Rebars and PT:
Gr60 for rebars;
ASTM A416-Gr270 for strands in PT tendons
Concrete:
f 'c = 45 Mpa, from 28 day concrete cylinder tests
For the concrete core wall channels:
the clear cover is 5/16"; the max. aggregate size is 1/4".
For all mass blocks and the foundation :
the clear cover is 1 3/16" and the max. aggregate size is 3/4".

Typical Development Length of Reinforcment

#2 Rebars:

1/8" Wires:



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Smart Structures

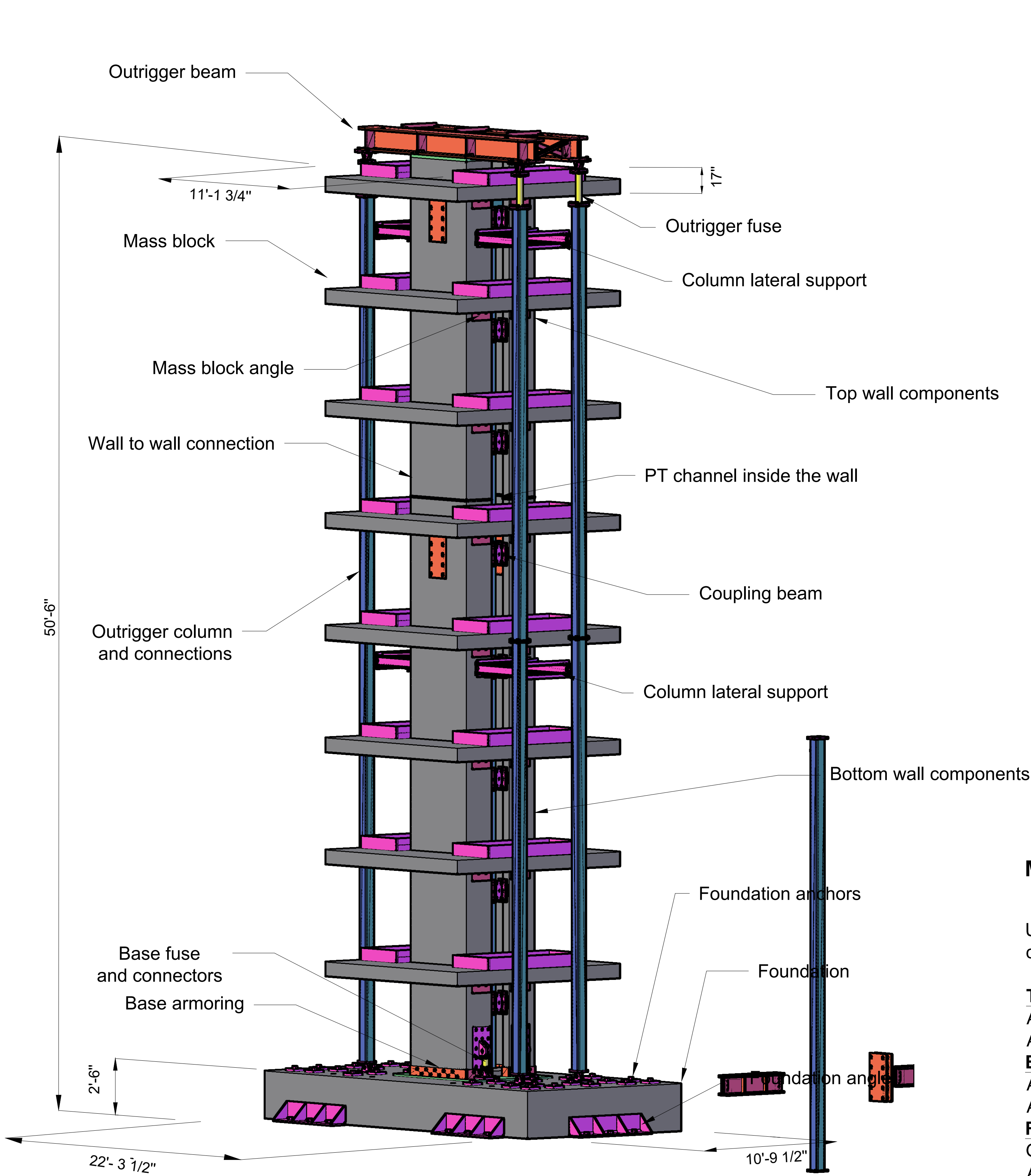
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

General Information

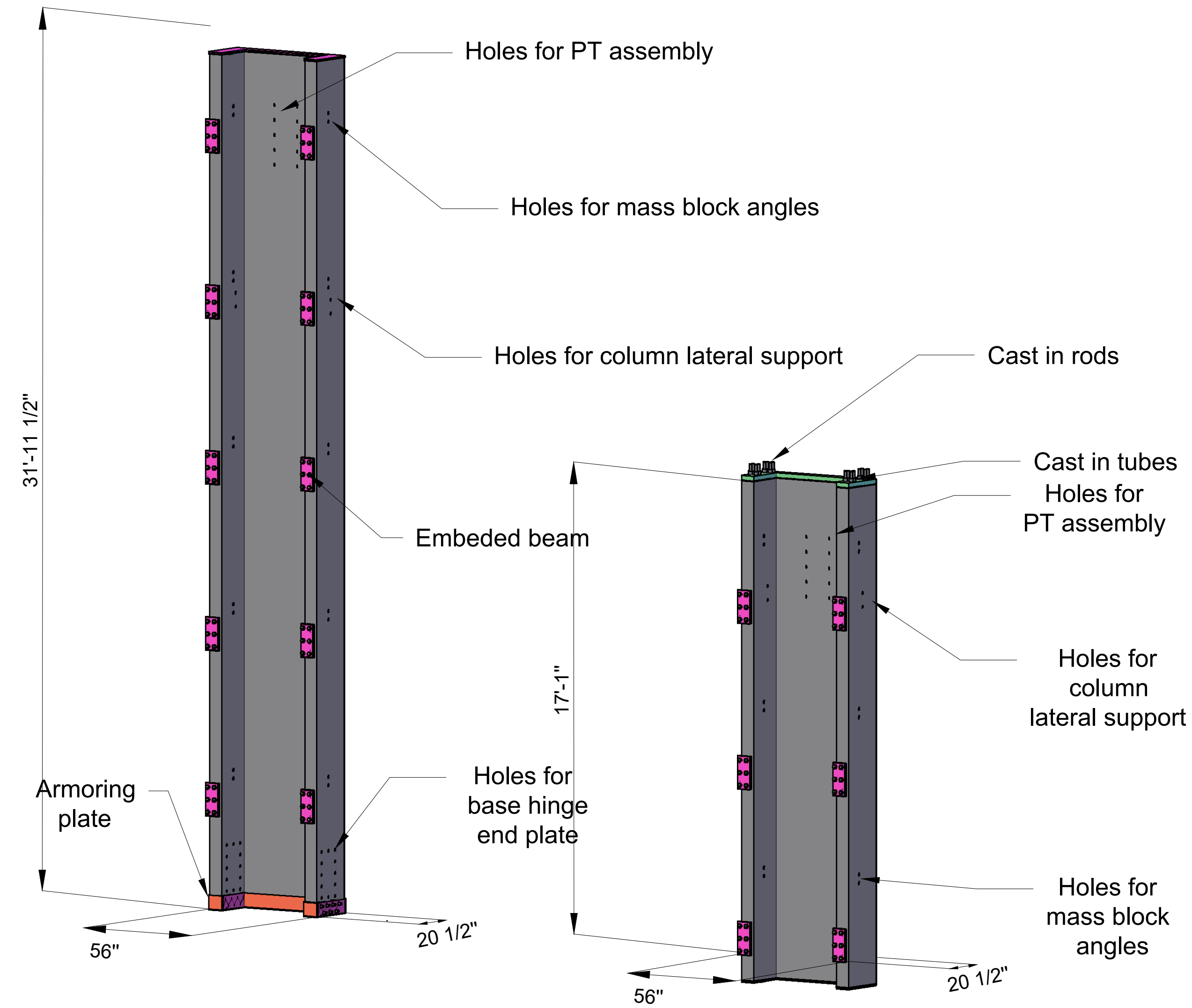
SHEET NO.
P01

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Shake table test



Bottom wall component
Qyt: 2
Unit Weight: 60kN/13.5kips

Top wall component
Qyt: 2
Unit Weight: 30kN/6.75kips

Material Properties

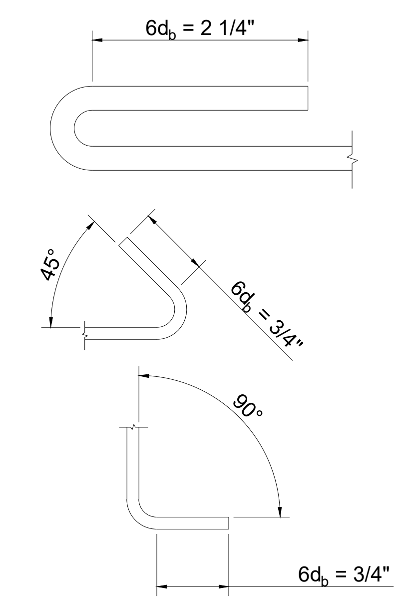
Unless otherwise specified in the drawings, materials for all components should be:

Threaded Rods:
ASTM F1554-Gr105 for threaded rods (no welding required);
ASTM F1554-Gr55 for threaded rods (welding required);
Bolts:
ASTM A490 for bolts to connect coupling beams
ASTM A325 for other bolts
Rebars and PT:
Gr60 for rebars;
ASTM A416-Gr270 for strands in PT tendons
Concrete:
f 'c = 45 Mpa, from 28 day concrete cylinder tests
For the concrete core wall channels:
the clear cover is 5/16"; the max. aggregate size is 1/4".
For all mass blocks and the foundation :
the clear cover is 1 3/16" and the max. aggregate size is 3/4".

Typical Development Length of Reinforcment

#2 Rebars:

1/8" Wires:



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Smart Structures

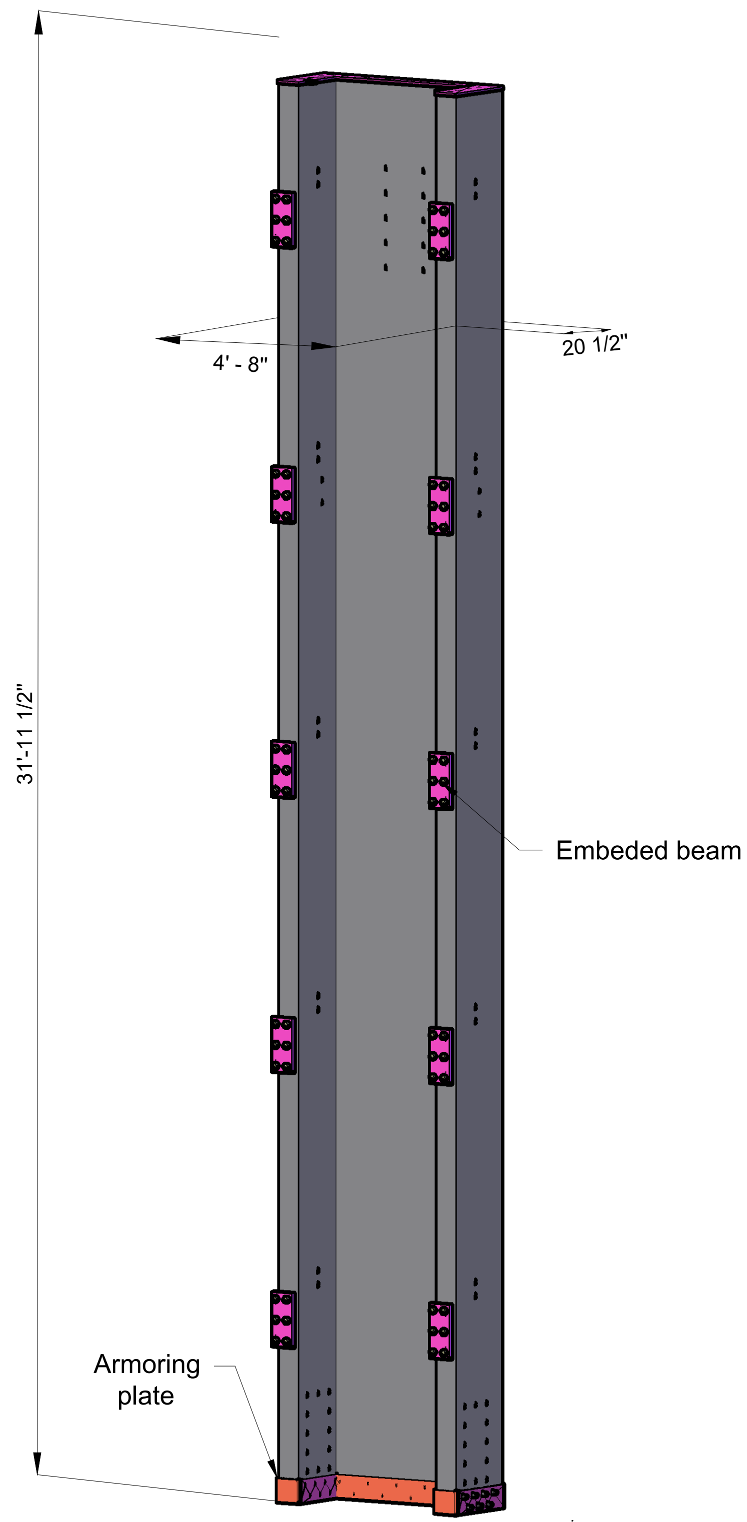
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

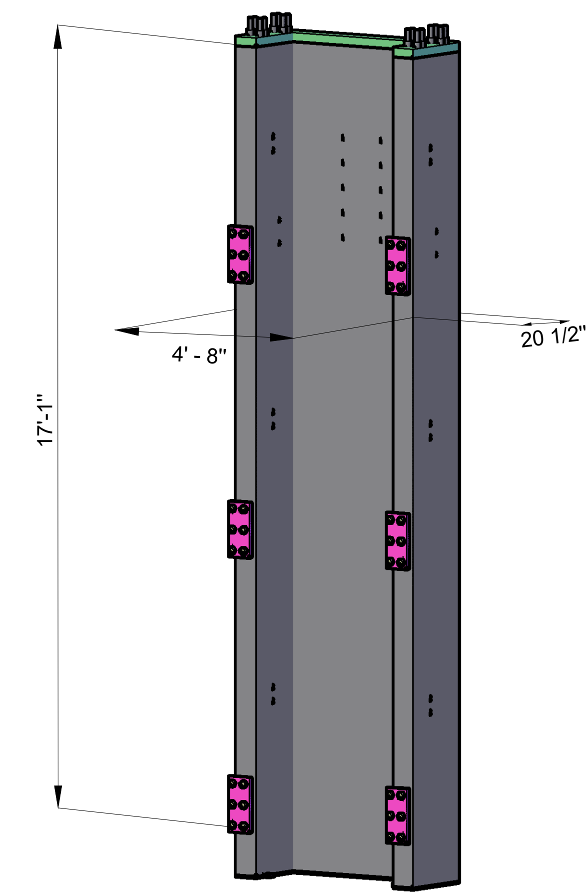
General Information

SHEET NO.
Cover

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08





Bottom wall component
Qyt: 2
Unit Weight: 60kN/13.5kips

Top wall component
Qyt: 2
Unit Weight: 30kN/6.75kips

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Smart Structures

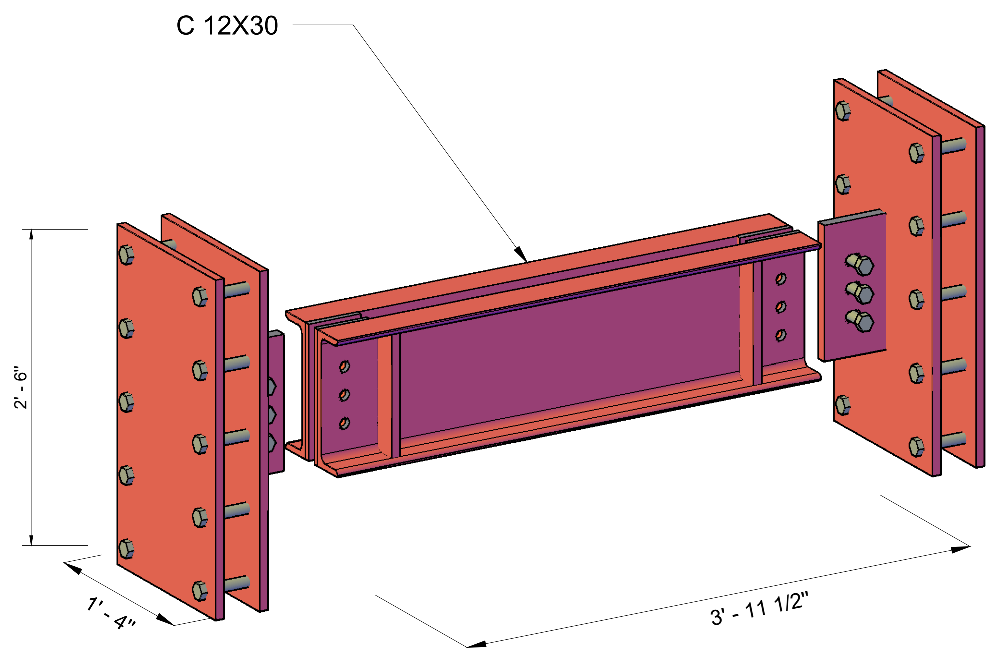
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Components in 3D View

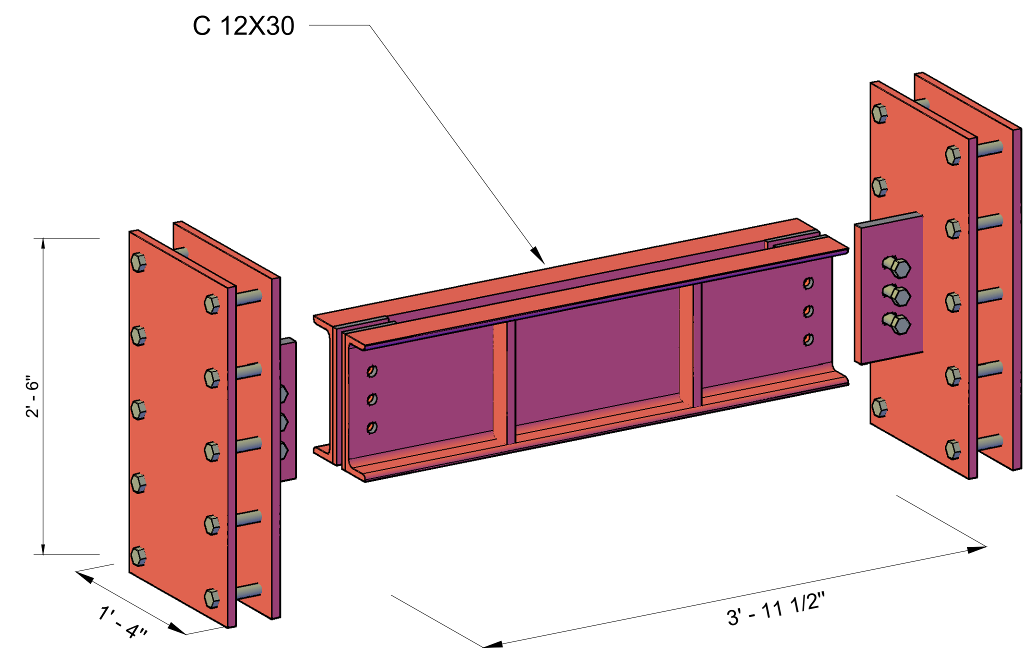
SHEET NO. 1

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

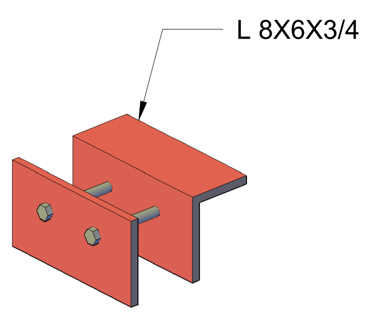
DATE 2023/10/08



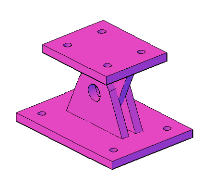
PT anchor - Type I
Qyt: 1
Unit Weight: 3.5kN/0.8kips



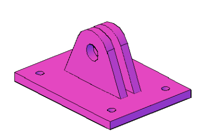
PT anchor - Type II
Qyt: 1
Unit Weight: 3.5kN/0.8kips



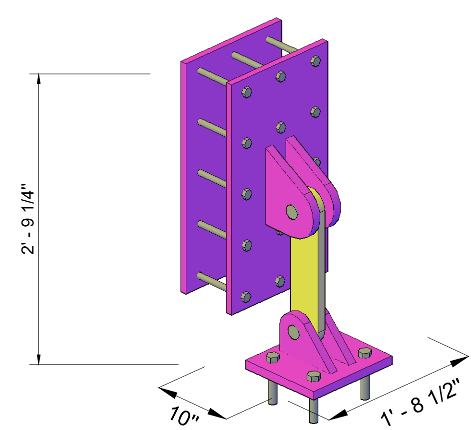
Mass block support
Qyt: 32
Unit Weight: 0.3kN/0.065kips



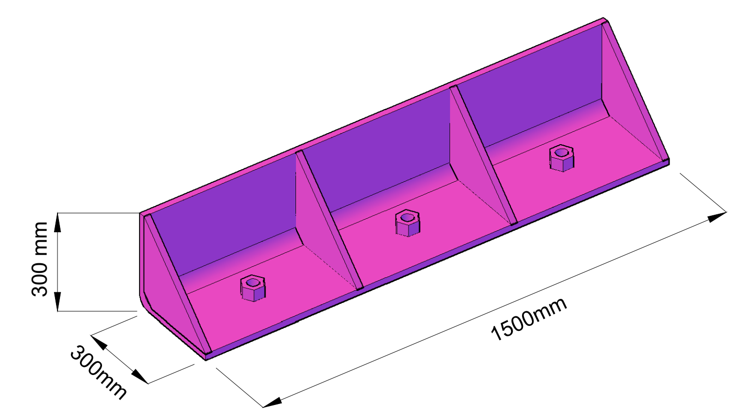
Outrigger end connection
Qyt: 4
Unit Weight: 0.2kN/0.04kips



Outrigger end connection
Qyt: 4
Unit Weight: 0.14kN/0.03kips



Base fuse
Qyt: 4
Unit Weight (without fuse): 1.1kN/0.25kips



Foundation shear angle (Formed by 3/4" Thk plate)
Qyt: 6
Unit Weight: 2.4kN/0.54kips

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

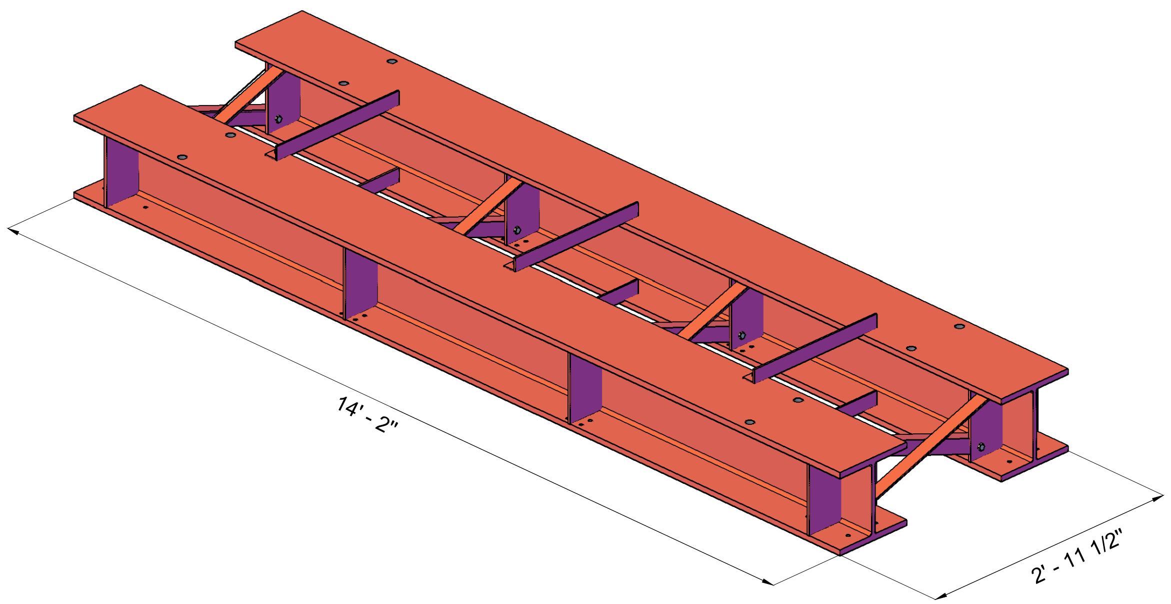
Components in 3D View

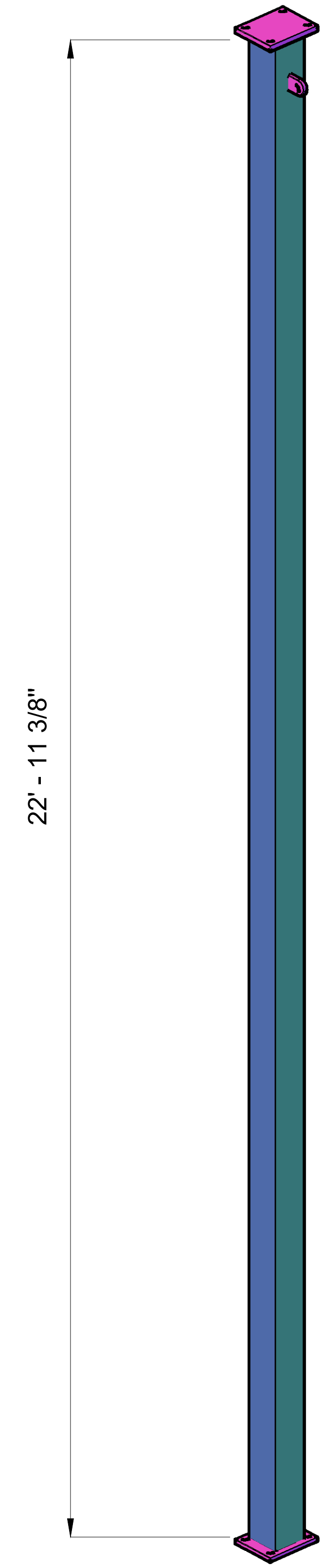
SHEET NO.
2

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

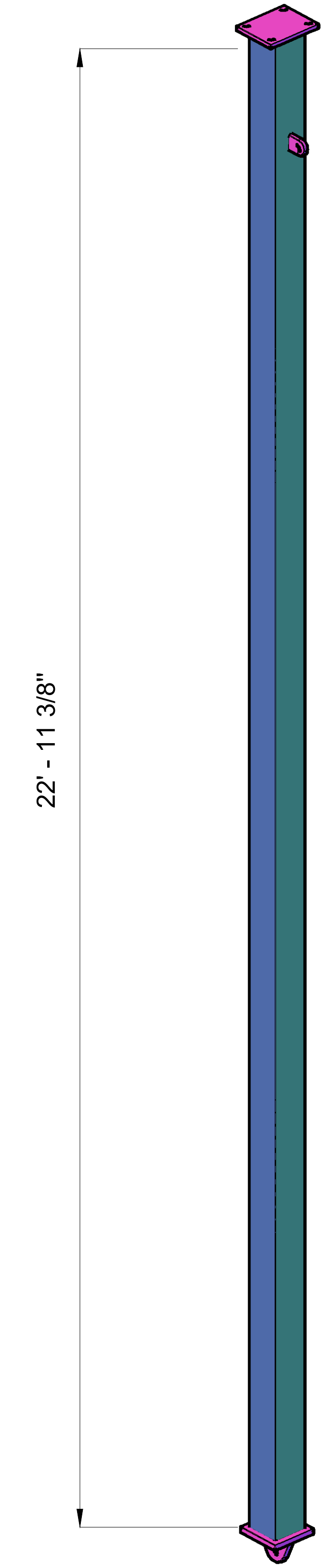
DATE 2023/10/08

Outrigger
Qyt: 1
Unit Weight: 16.8kN/3.8kips

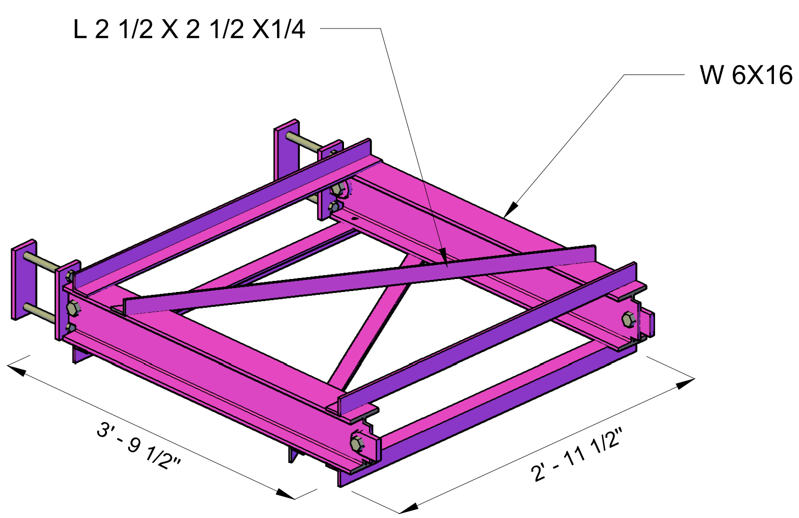




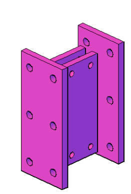
Outrigger column - Type I
Qyt: 4
Unit Weight: 3.1kN/0.7kips



Outrigger column - Type II
Qyt: 4
Unit Weight: 3.1kN/0.7kips



Column lateral support
Qyt: 4
Unit Weight: 0.75kN/0.17kips



Friction Device
Qyt: 16
Unit Weight: 0.2kN/0.04kips

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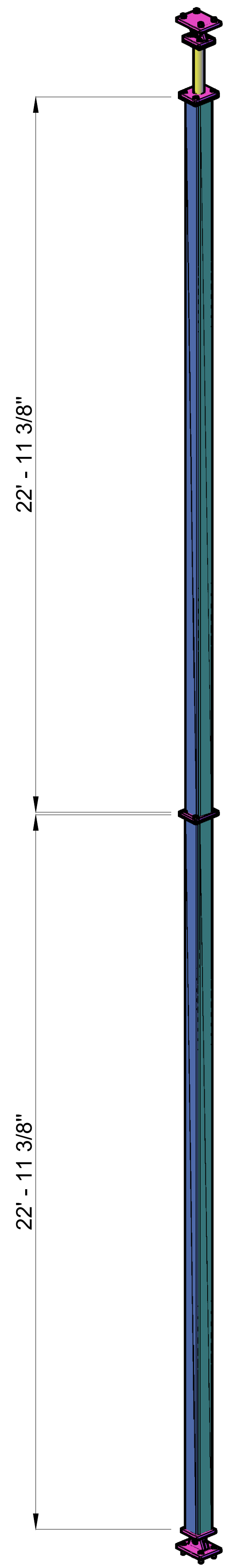
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

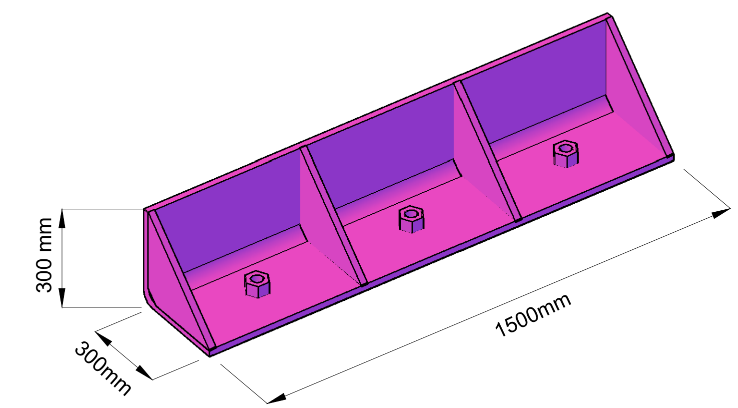
Components in 3D View

SHEET NO.
3

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08





Foundation shear angle (Formed by 3/4" Thk plate)
Qyt: 6
Unit Weight: 2.4kN/0.54kips

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Components in 3D View

SHEET NO.
4

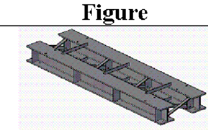
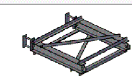
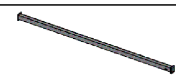
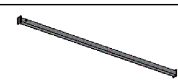
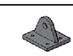
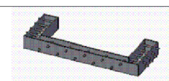
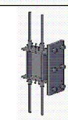
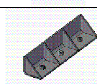
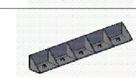
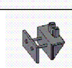
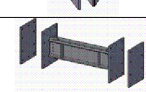
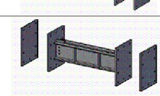
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

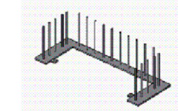
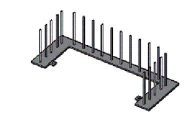
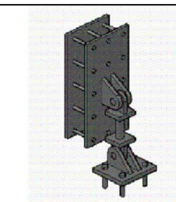
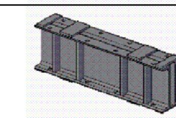
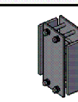
DATE 2023/10/08

Steel Components List

Manufactured by GTS

Ready for Fabrication

Num	Component	Figure	Qty	Drawing
1	Outrigger		1	**S1-1**
2	Outrigger column lateral support		4	**S2**
3	Outrigger columnc Type I		4	**S3-1**
4	Outrigger columnc Type II		4	**S3-1**
5	Outrigger end connection Type I		8	**S3-2**
6	Outrigger end connection Type II		4	**S3-2**
7	Wall base armoring		2	**S4-1/S4-2**
8	Steel embedded beam		32	**S5**
9	Foundation anchor Type I		4	**S6-1**
10	Foundation anchor Type II		2	**S6-2**
11	Mass block support		32	**S7**
12	PT anchor Type I		1	**S8-1/S8-3**
13	PT anchor Type II		1	**S8-2/S8-3**

Num	Component	Figure	Qty	Drawing
14	Wall to wall connection Type I		2	**S9**
15	Wall to wall connection Type II		2	**S9**
16	Base fuse connection		4	**S10**
17	PT base anchor plate		1	**S11**
18	Friction damper		16	**S12 UnderTest**

Standard Section Used

Num	Section Type	Total Length
1	W 14X120	28'-2"
2	W 8X15	24'-8''
3	W 6X16	30'-4''
4	L 8X6X3/4	32'
5	L 2.5X2.5X1/4	61'-3 3/4"
6	HSS 6X6X3/8	45'-10 3/4"
7	C12X30	22'-3''

Material Properties

Steel:

ASTM A36 for steel plates;

G40.21 300W for angles;

ASTM A572- Gr50 for hot-rolled W, M, HP, and channel sections

ASTM A500 - Gr50 for HSS sections

Threaded Rods:

ASTM F1554-Gr105 for threaded rods (no welding required);

ASTM F1554-Gr55 for threaded rods (welding required);

Bolts:

ASTM A325 for bolts

THE UNIVERSITY OF BRITISH COLUMBIA

Smart Structures

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

General Information

SHEET NO.

Cover

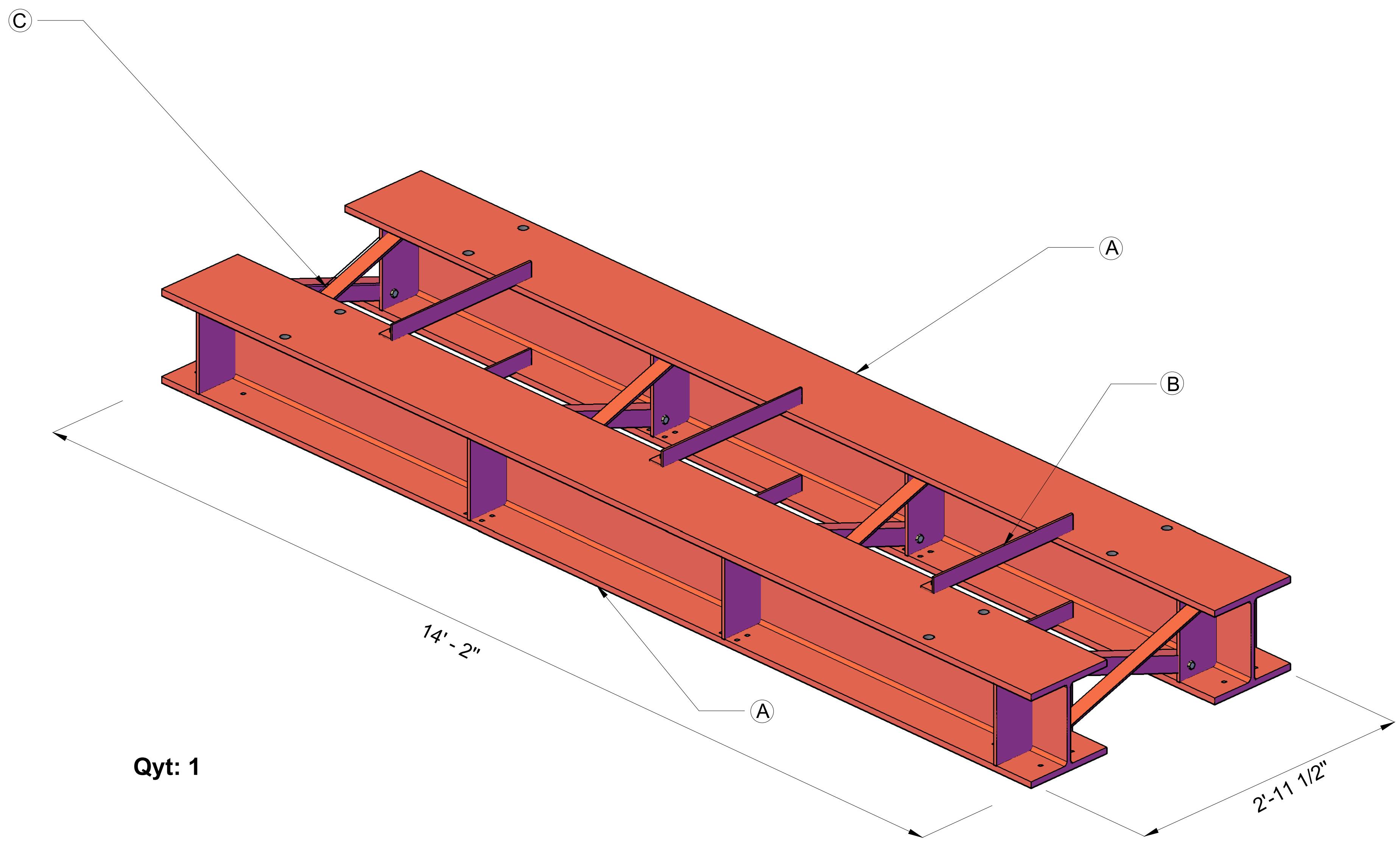
DRAWN BY

Tianyang Qiao

Hengchao Xu

Lisa Tobber

DATE 2023/10/08



Qyt: 1

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Smart Structures

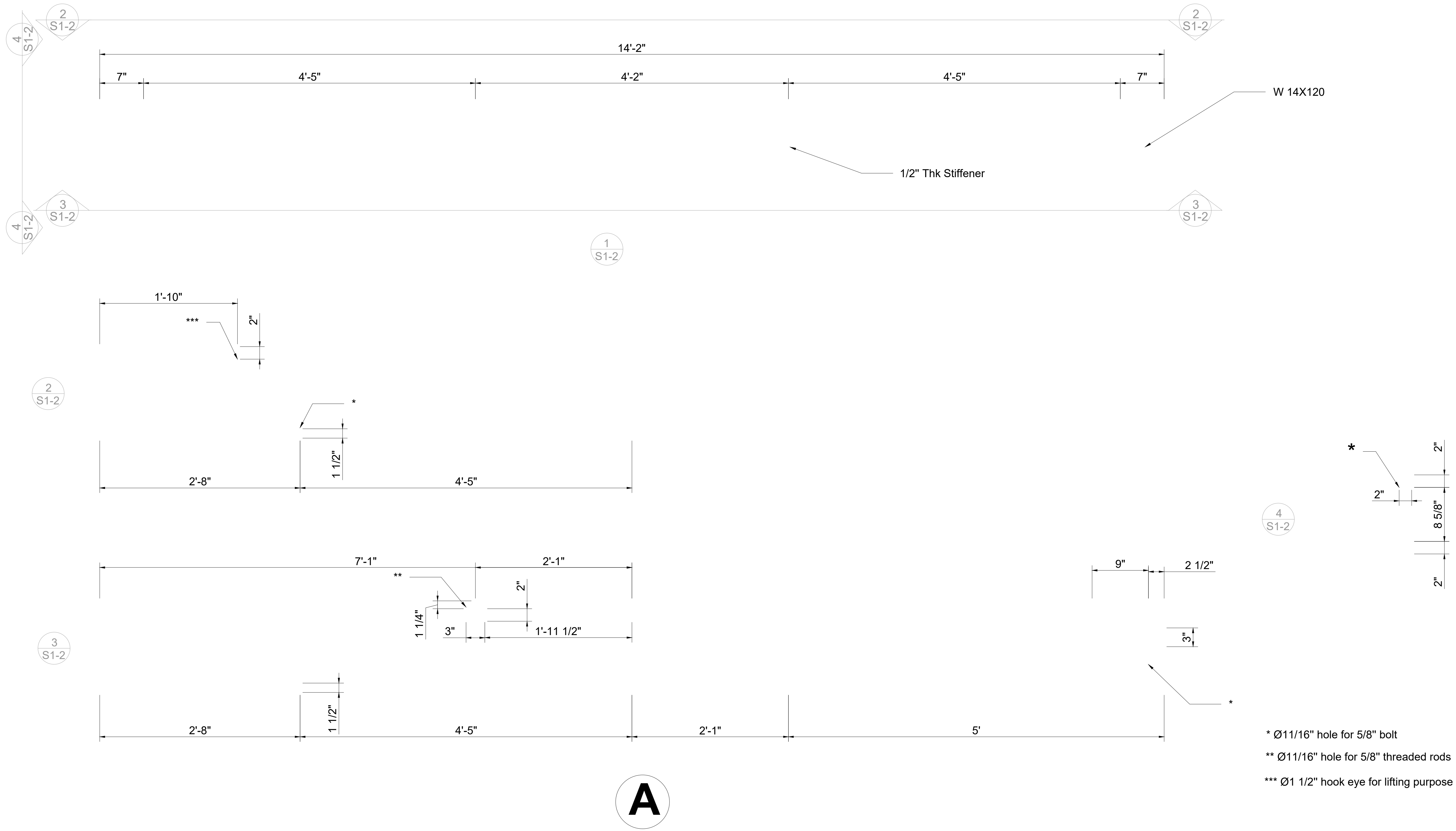
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Outrigger

SHEET NO.
S1-1

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

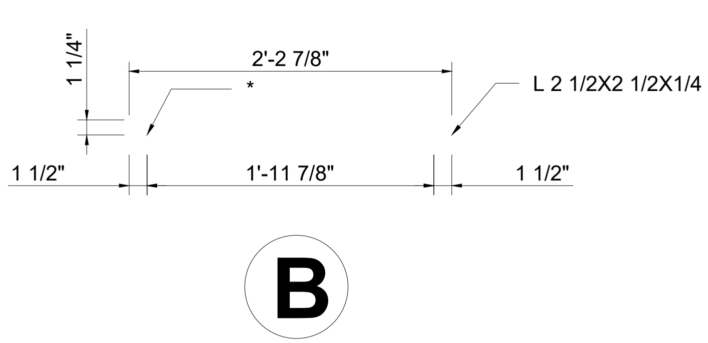
DATE	2023/10/08

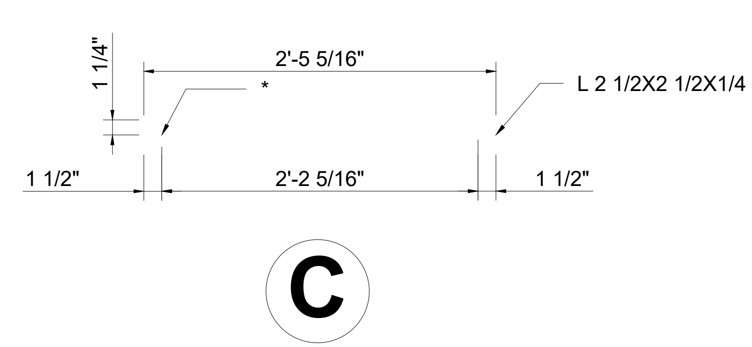


* Ø11/16" hole for 5/8" bolt

** Ø11/16" hole for 5/8" threaded rods

*** Ø1 1/2" hook eye for lifting purpose





* Ø11/16" hole for 5/8" bolt

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Smart Structures

ILEE

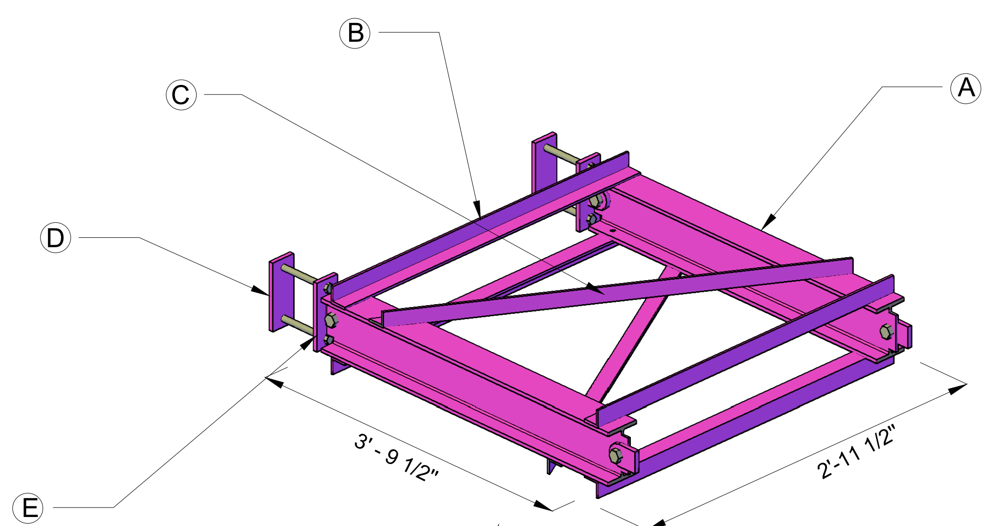
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Outrigger

SHEET NO.
S1-2

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08

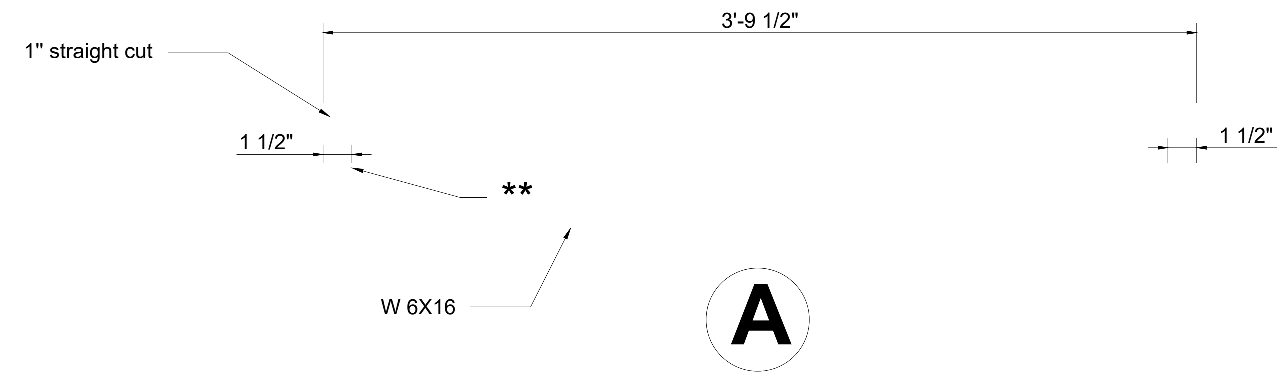


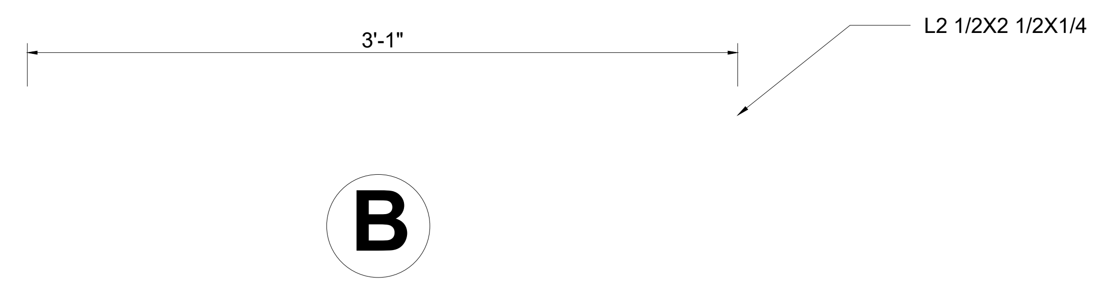
Qyt: 4

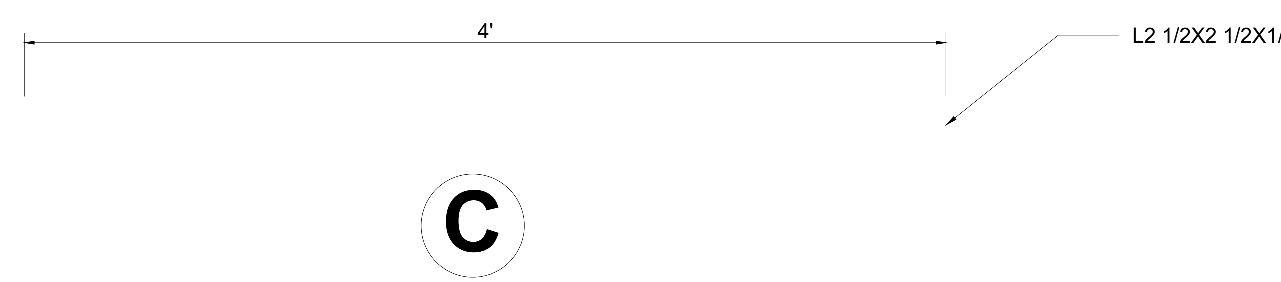
Notes for GTS:

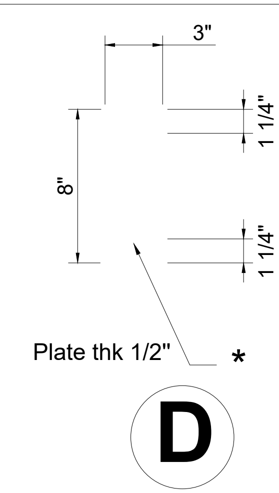
The purpose of this column lateral support is to provide stability to outrigger columns.

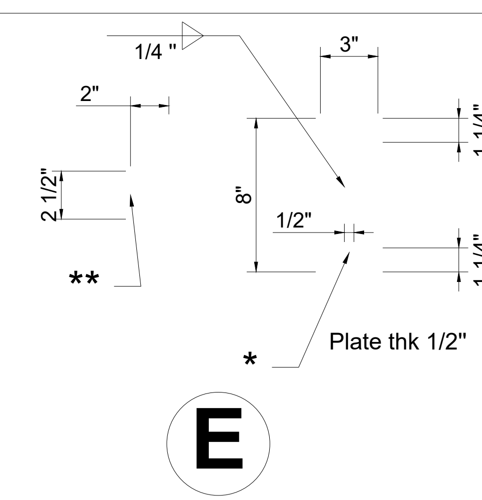
Currently, as shown in the figure, there are three angles welded at the top of the I-beam and another three welded at the bottom of the I-beam. They don't have to be in this exact configuration. So, if there is an easier way to construct this, please let us know.











* Ø11/16" hole for 5/8" bolt

** Drill Ø15/16" hole for 7/8" bolt

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Smart Structures

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Outrigger Column Lateral Support

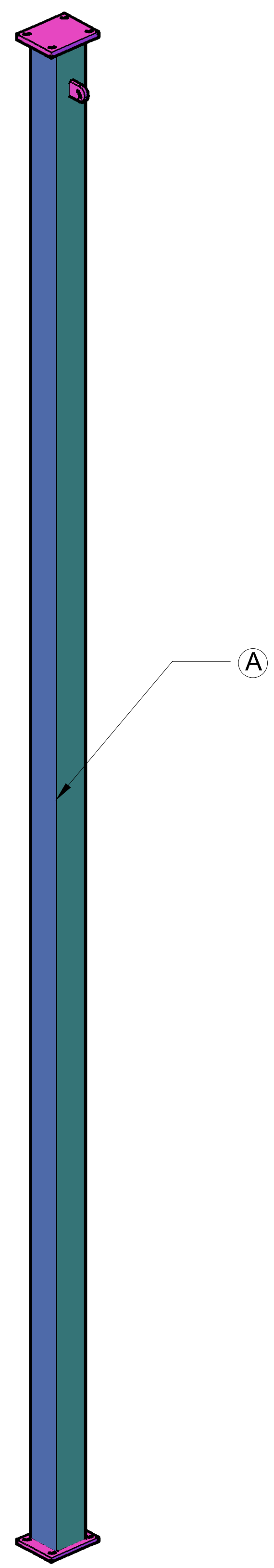
SHEET NO.

S2

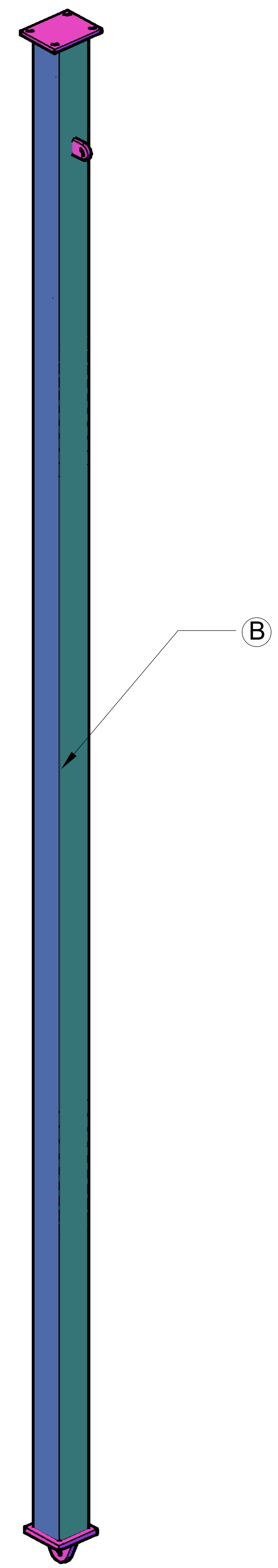
DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

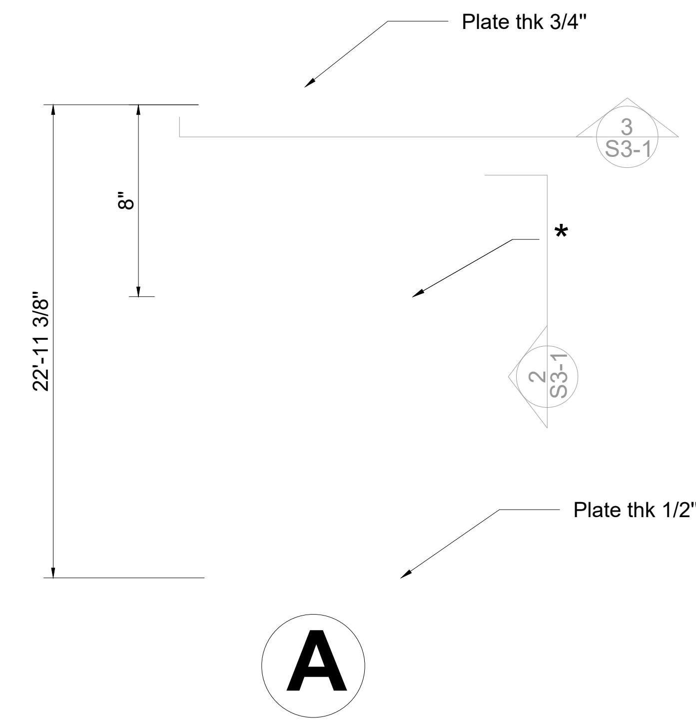
DATE 2023/10/08

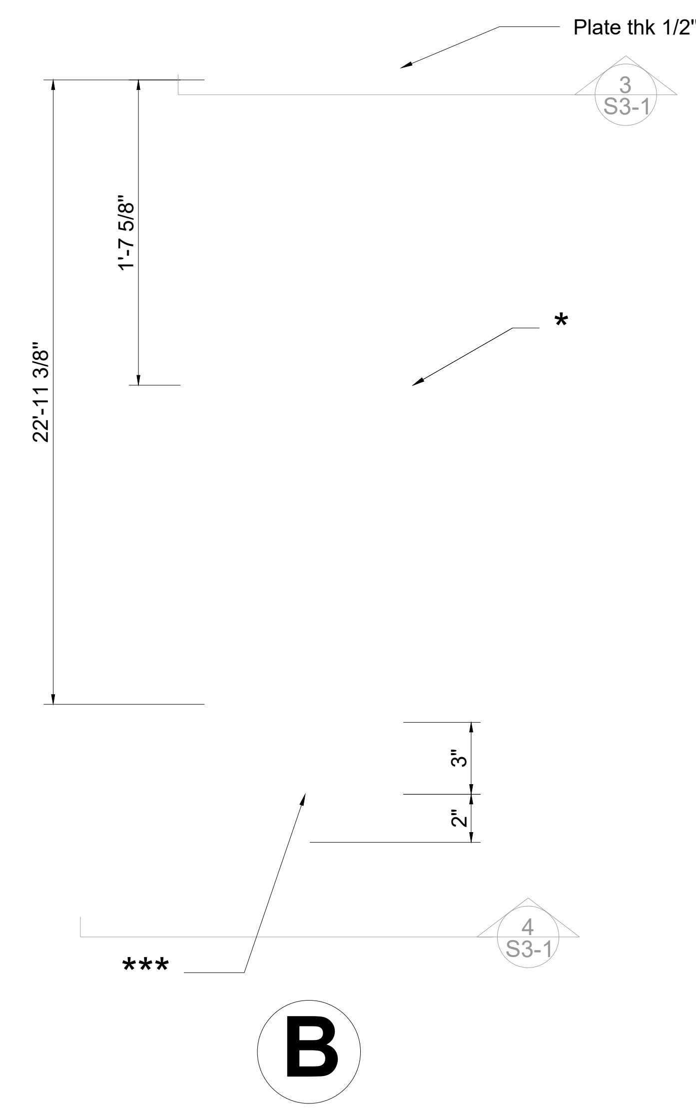


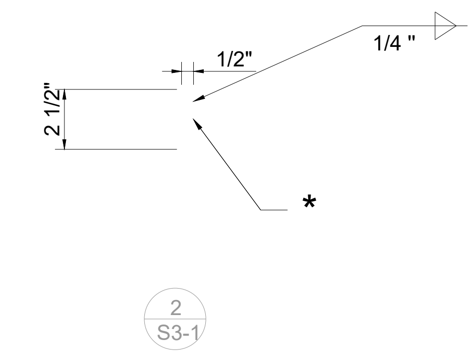
HSS 6X6X3/8
Qyt: 4

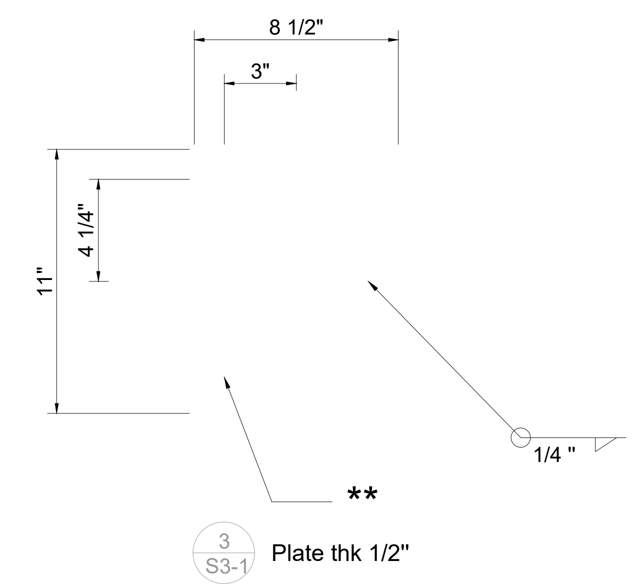


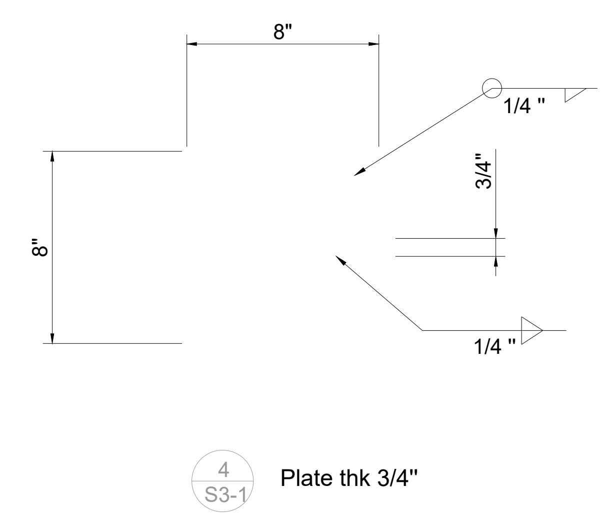
HSS 6X6X3/8
Qyt: 4











* Ø15/16" hole for 7/8" bolt
** Ø11/16" hole for 5/8" bolts
*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

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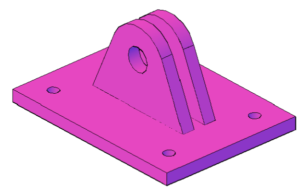
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Outrigger Column

SHEET NO.
S3-1

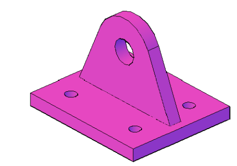
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



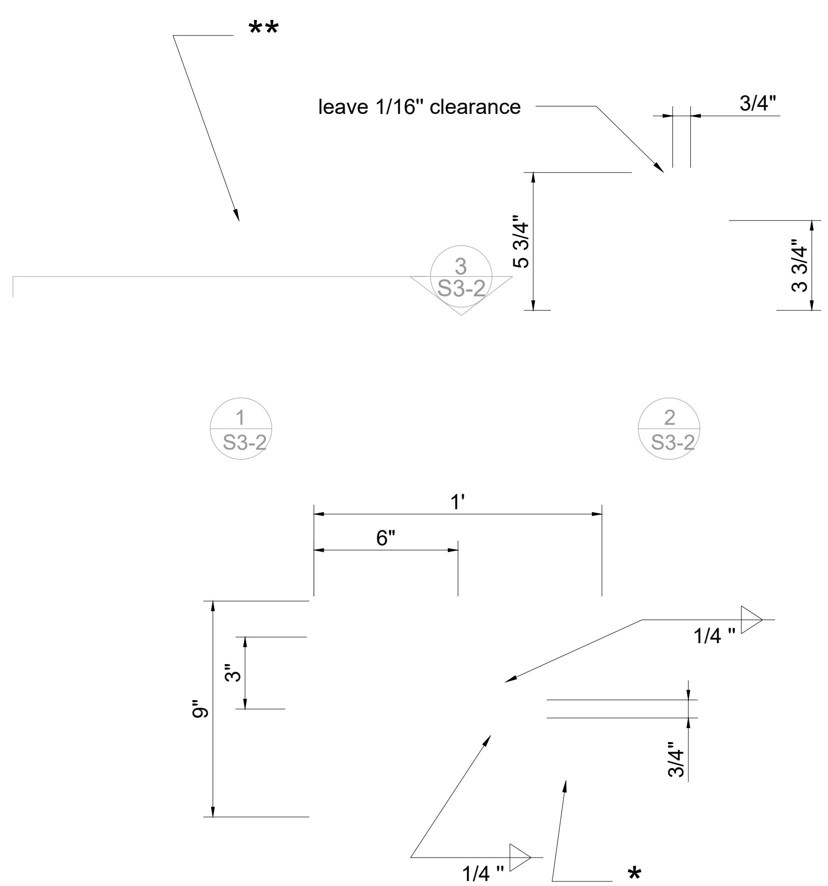
Type I

Qty: 8



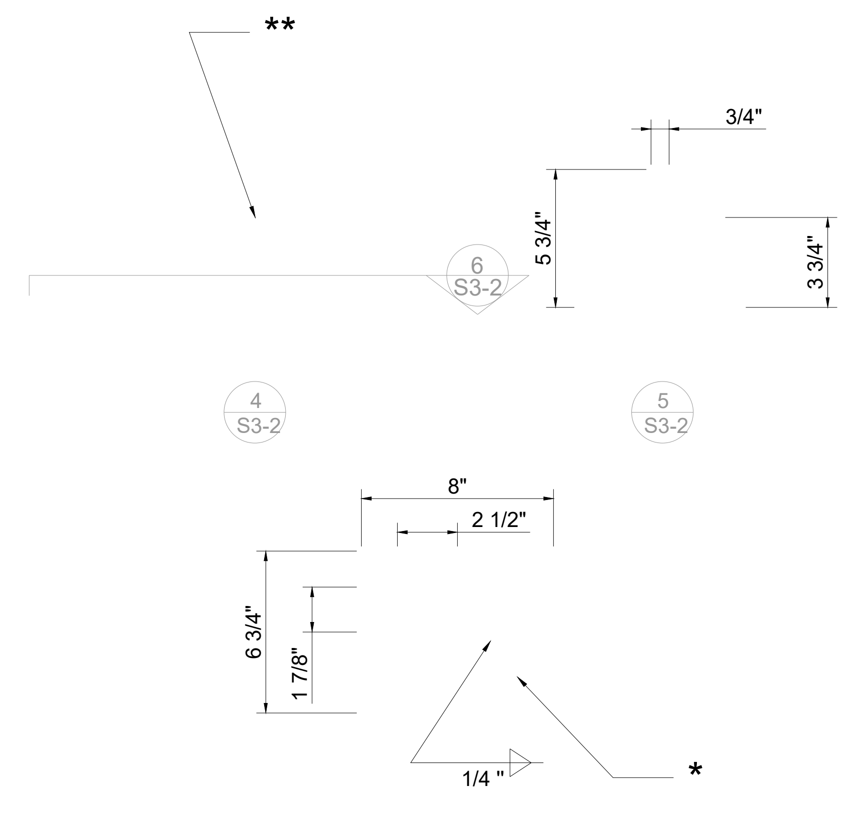
Type II

Qty: 4



3 S3-2 Plate thk 3/4"

Type I



6 S3-2 Plate thk 3/4"

Type II

* Ø11/16" hole for 5/8" threaded rods

** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Outrigger End Connection

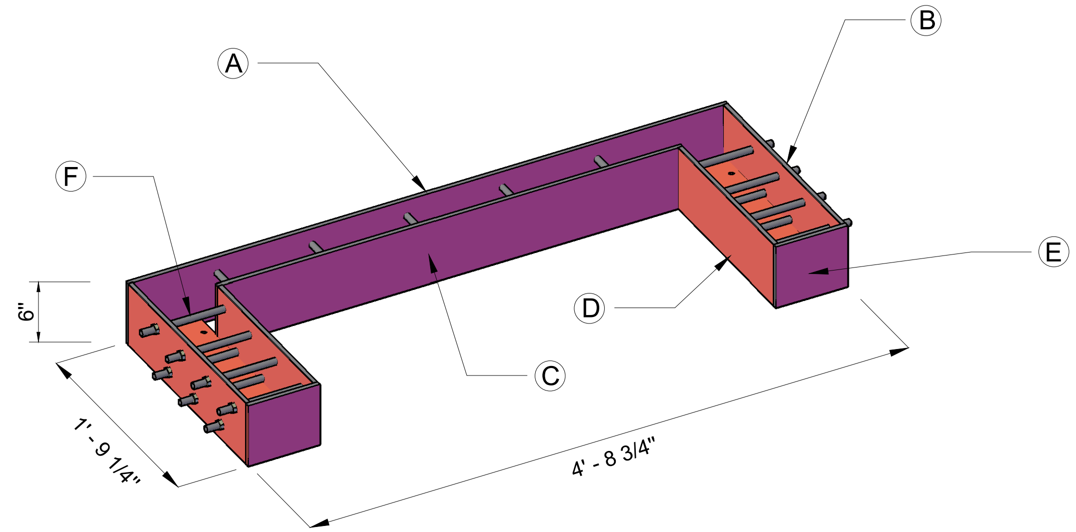
SHEET NO.

S3-2

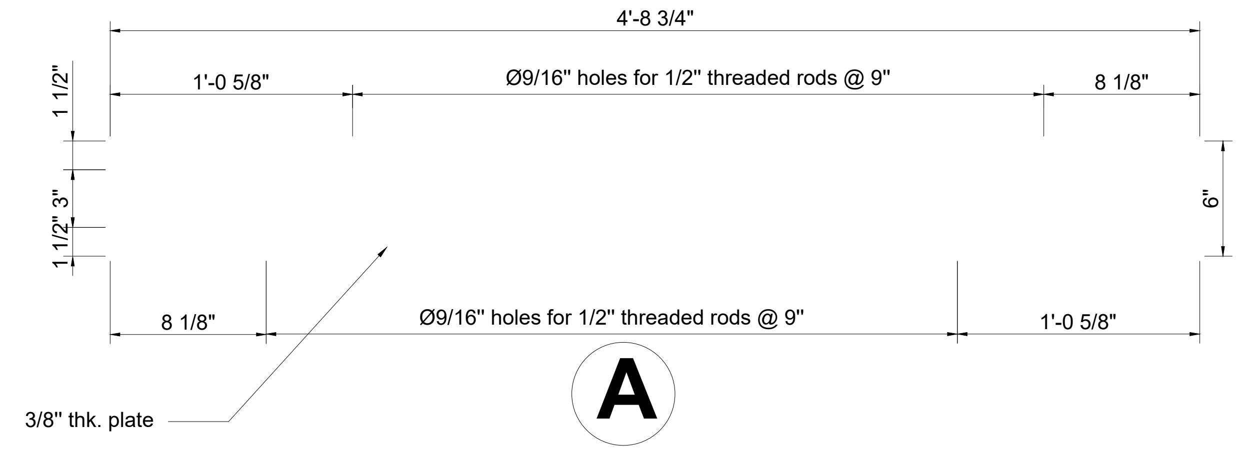
DRAWN BY

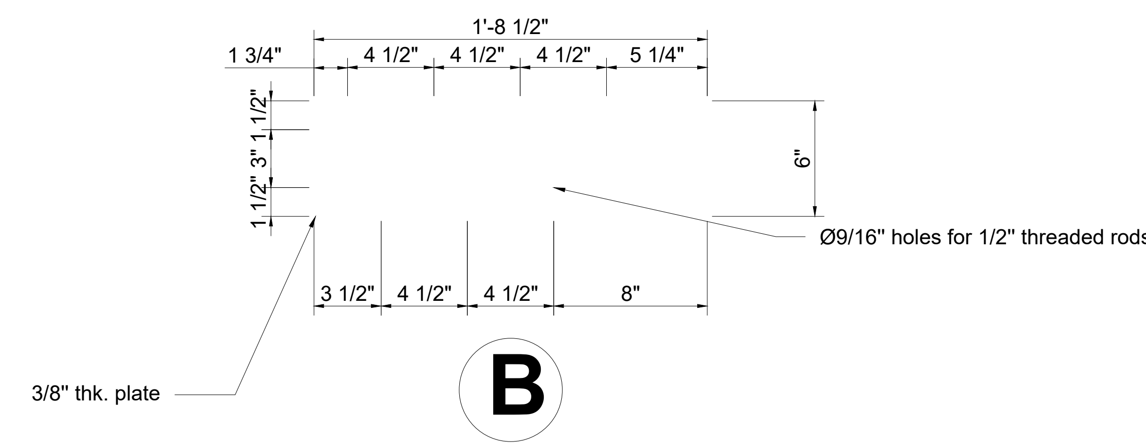
Tianyang Qiao
Hengchao Xu
Lisa Tobber

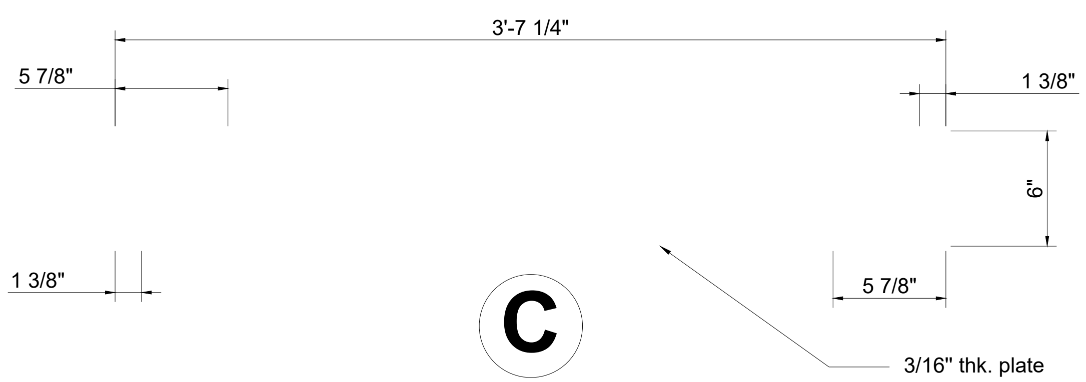
DATE 2023/10/08

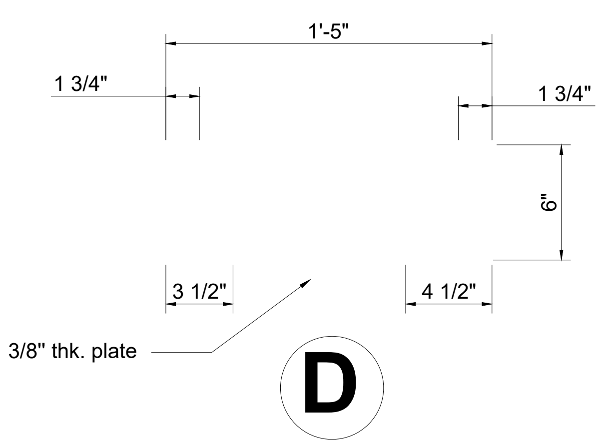


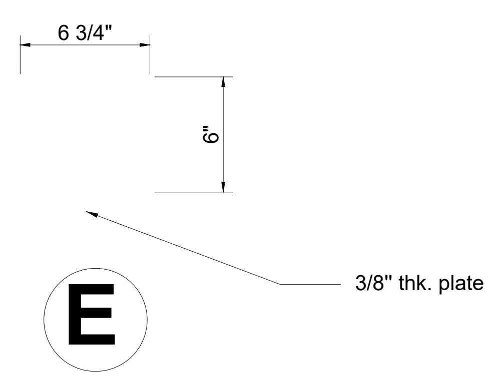
Qyt: 2

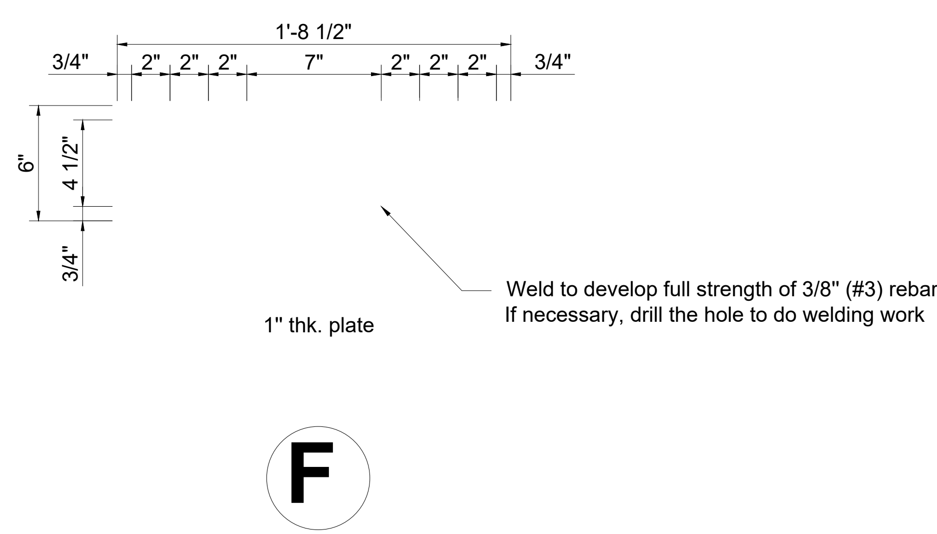












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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Wall Base Armoring

SHEET NO.

S4-1

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08

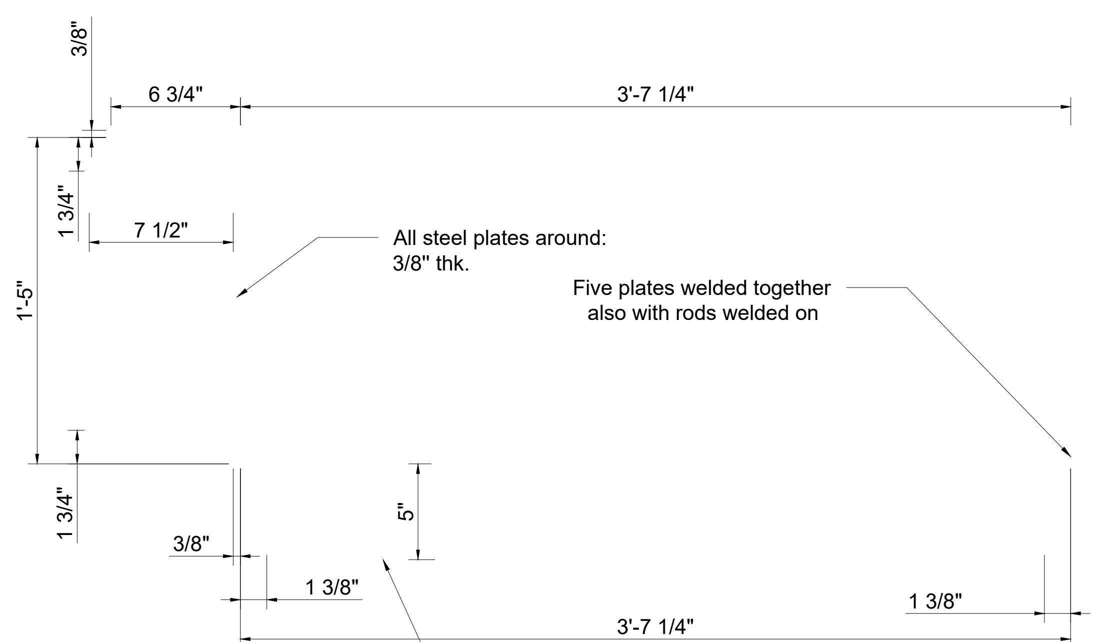
1" thk plate with holes for rebar
is welded to armoring plate, covering the flange from bottom

All steel plates around:
3/8'" thk.

All threaded rods: welded to plate at one end (interior),
and bolted at the another end (assemble before casting concrete)

1
S4-2

Five 3/8" thk.steel plates
welded together to be a frame



All steel plates around:
3/8" thk.

Five plates welded together
also with rods welded on

2
S4-2

All ASTM F1554 3/8" welded threaded rods:
stick out the form

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ILEE

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Wall Base Armoring

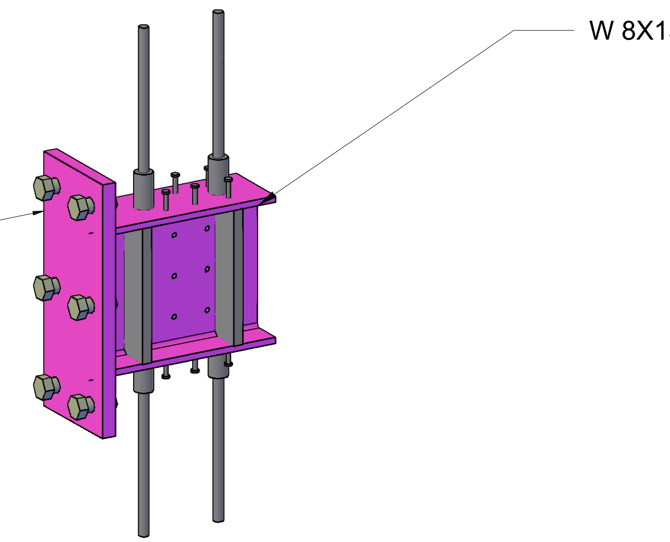
SHEET NO.
S4-2

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

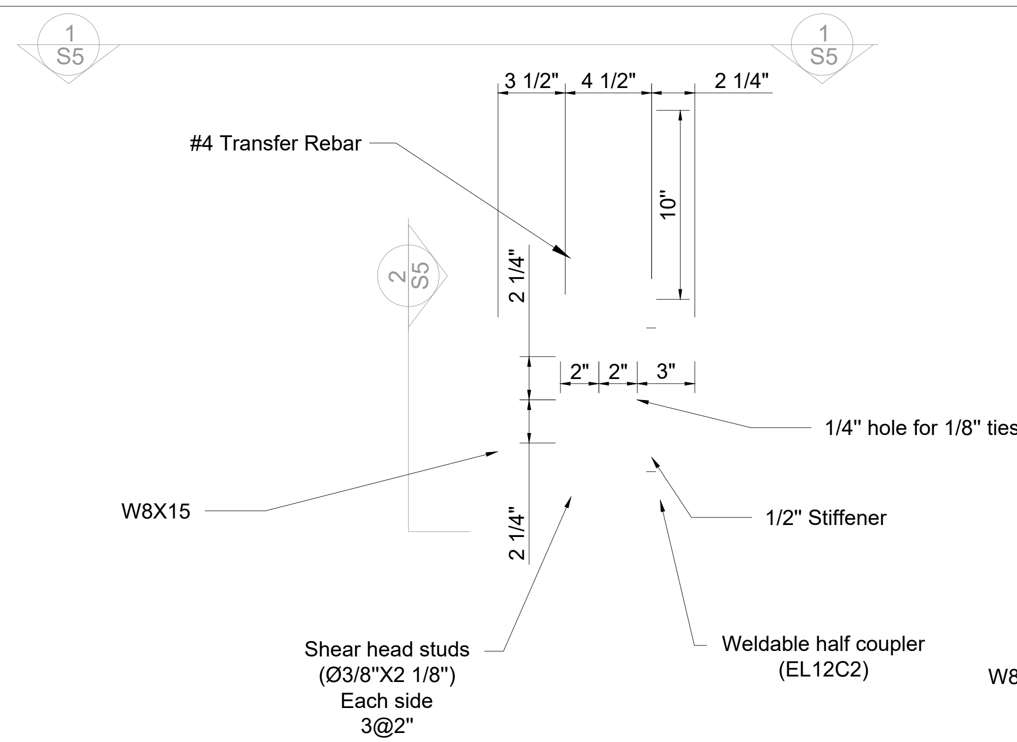
DATE 2023/10/08

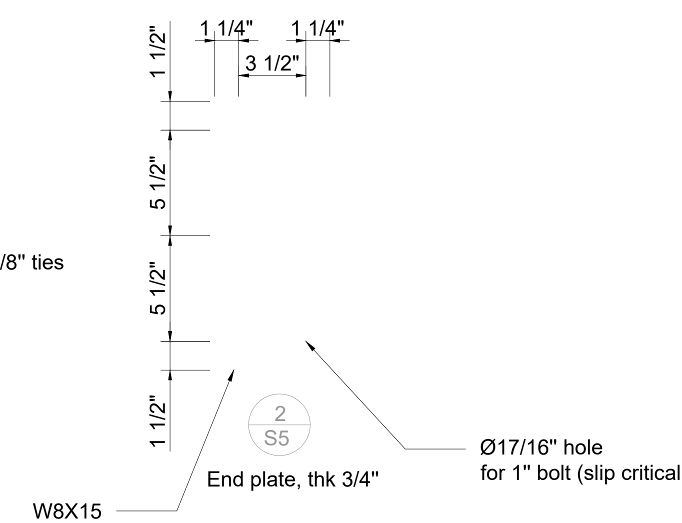
Notes for GTS:

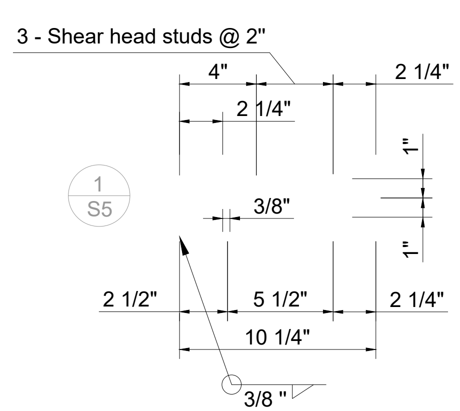
Based on the conversation with Rob, we need steel components to connect the beam to the form work. Please add as you see fit.



Qyt: 32







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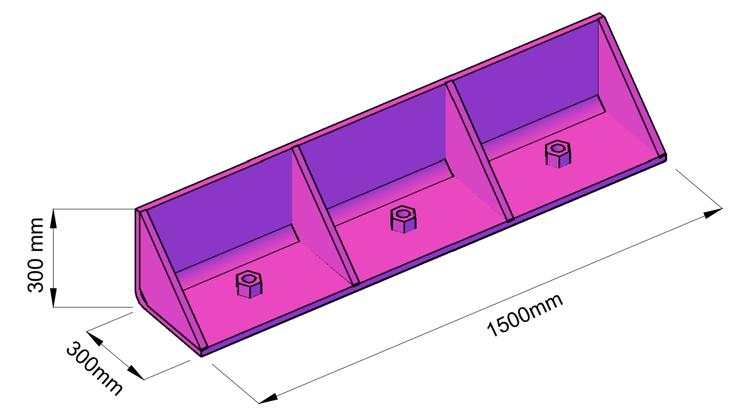
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

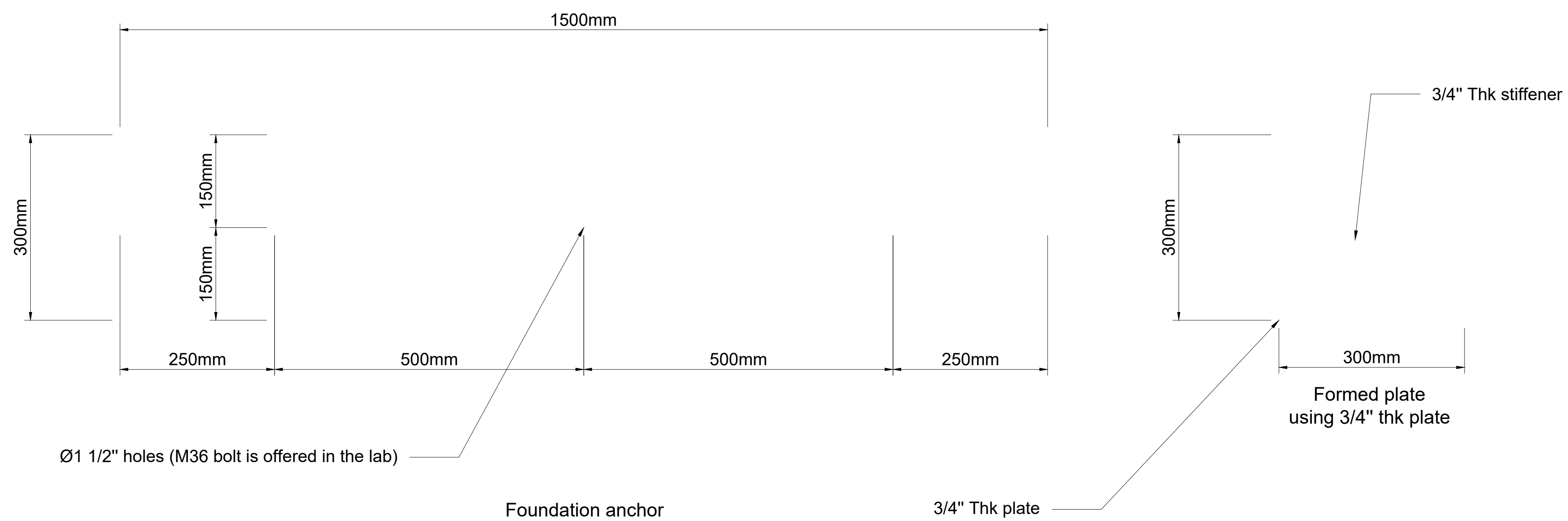
Steel Embedded Beam

SHEET NO.	**S5**
DRAWN BY	Tianyang Qiao Hengchao Xu Lisa Tobber
DATE	2023/10/08



Type I
Qyt: 6

Note: To match metric hole pattern in China, length and width are using metric unit, except the thickness, which is 3/4"



Foundation anchor

Qty. 4

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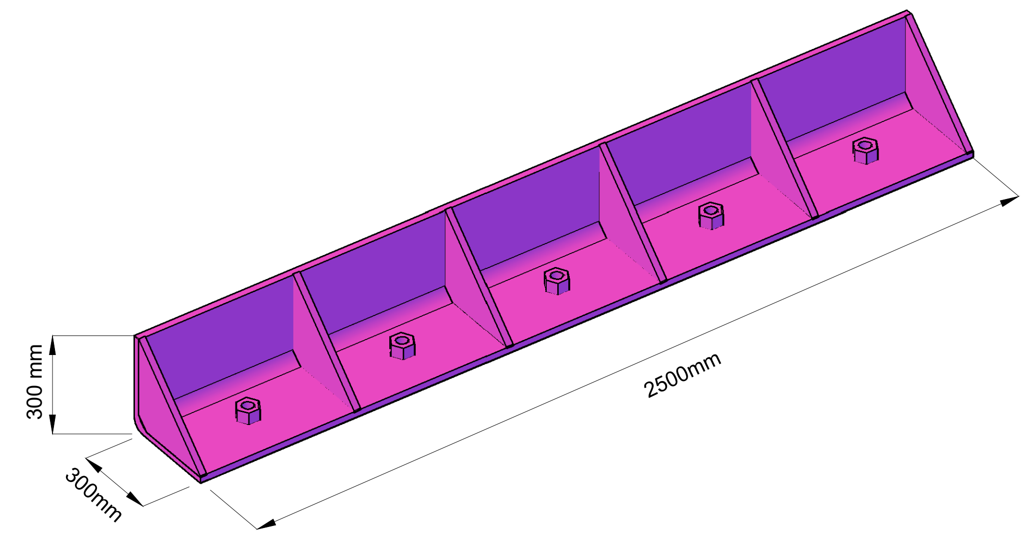
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Steel Embedded Beam

SHEET NO.
S6-1

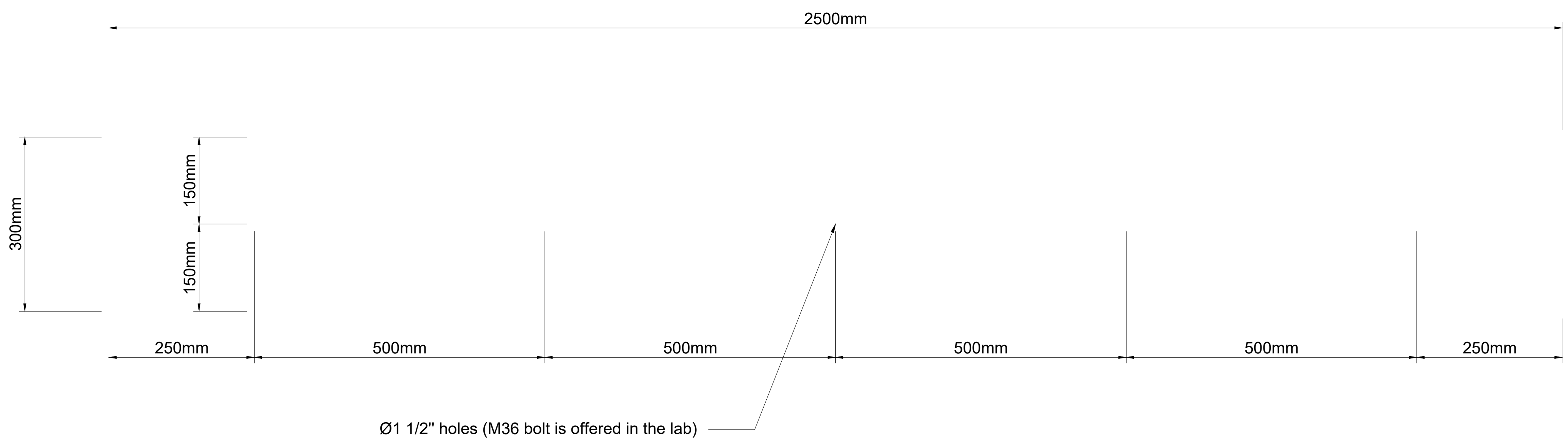
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



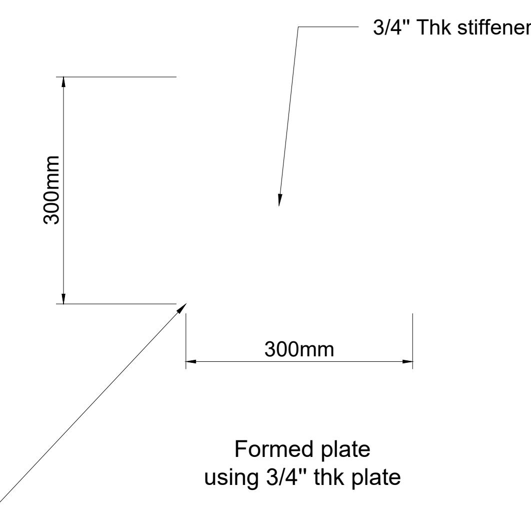
Type II
Qyt: 6

Note: To match metric hole pattern in China, length and width are using metric unit, except the thickness, which is 3/4"



Foundation anchor

Qty. 2



Formed plate
using 3/4" thk plate

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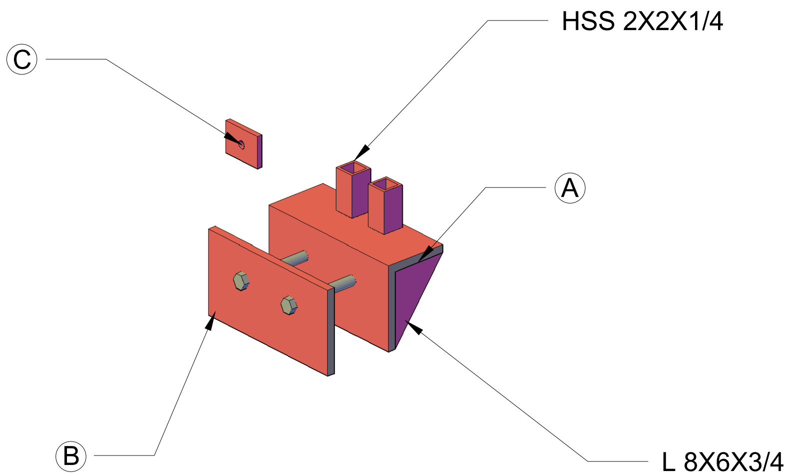
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Steel Embedded Beam

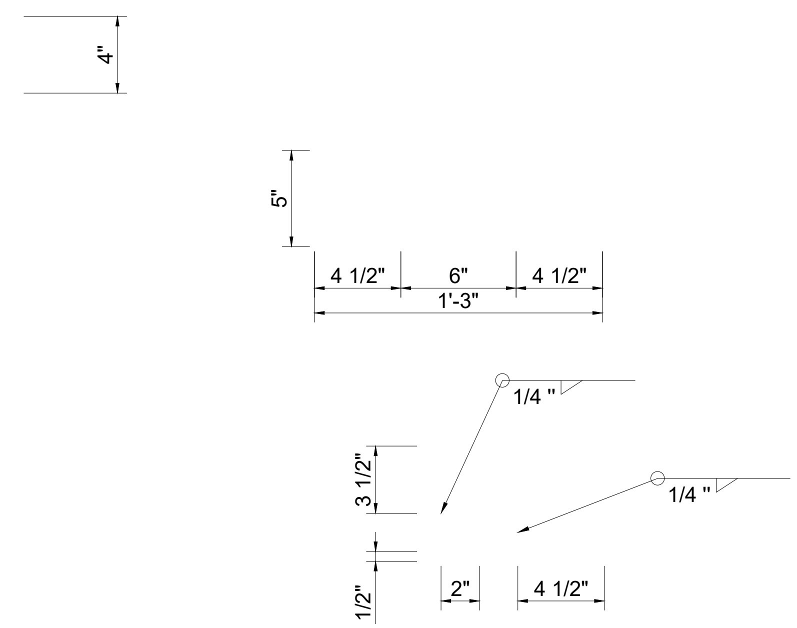
SHEET NO.
S6-2

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

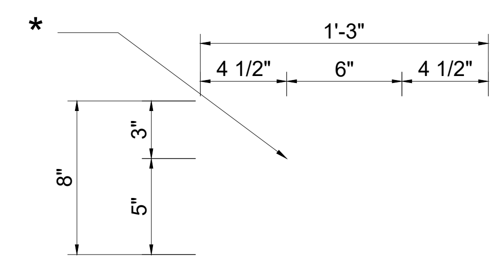
DATE 2023/10/08



Qyt: 32

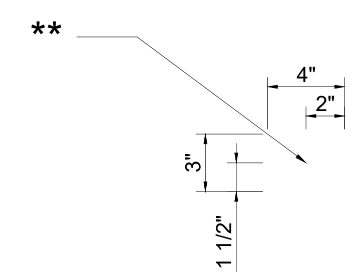


A



Thk 3/4" Plate

B



Thk 1/2" Plate, put in place during construction

C

* Ø15/16" hole for 7/8" bolts
** Ø13/16" hole for #6 rebar

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Smart Structures

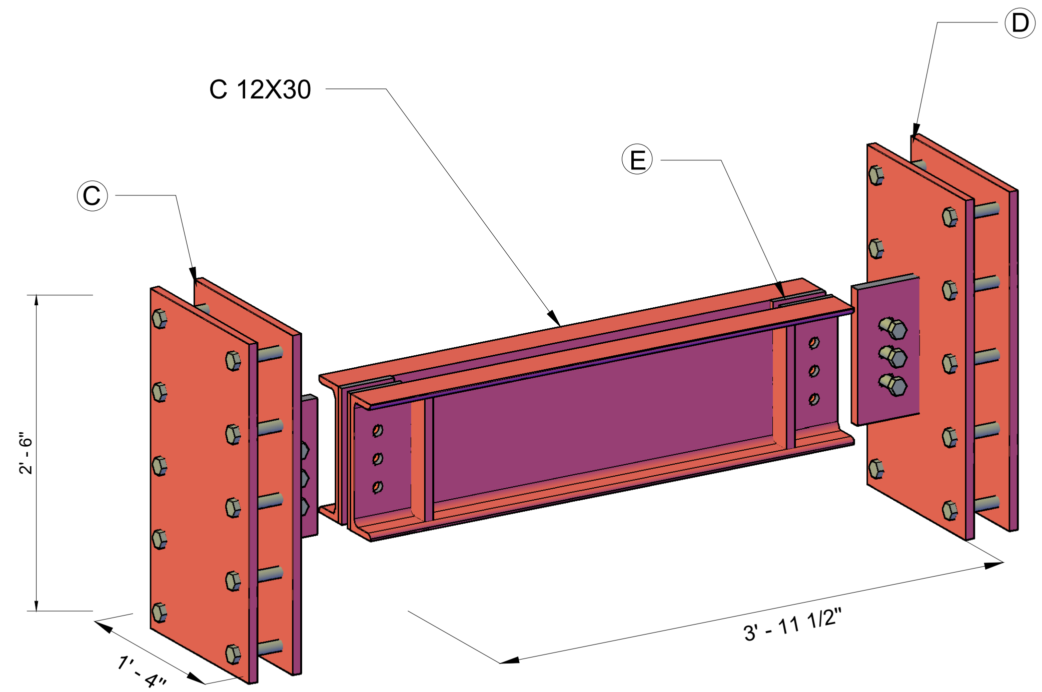
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

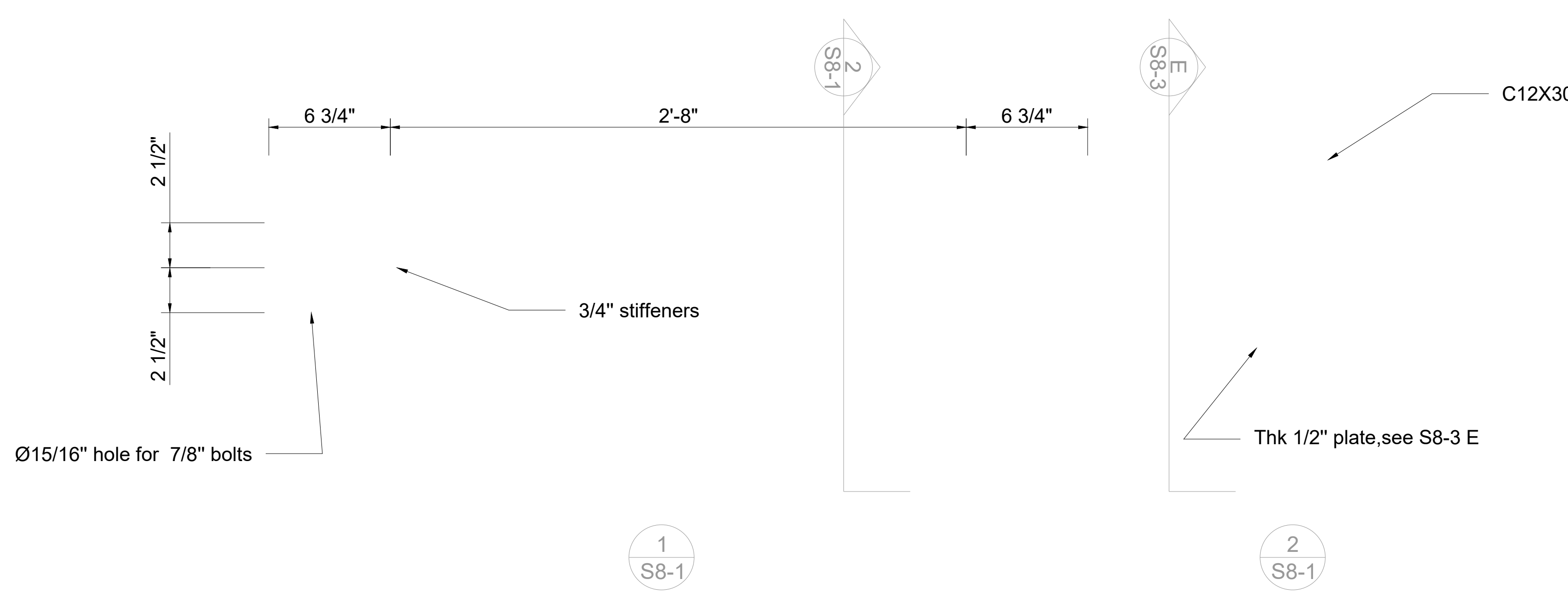
SHEET NO.
S7

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Type I
Qyt: 1



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Smart Structures

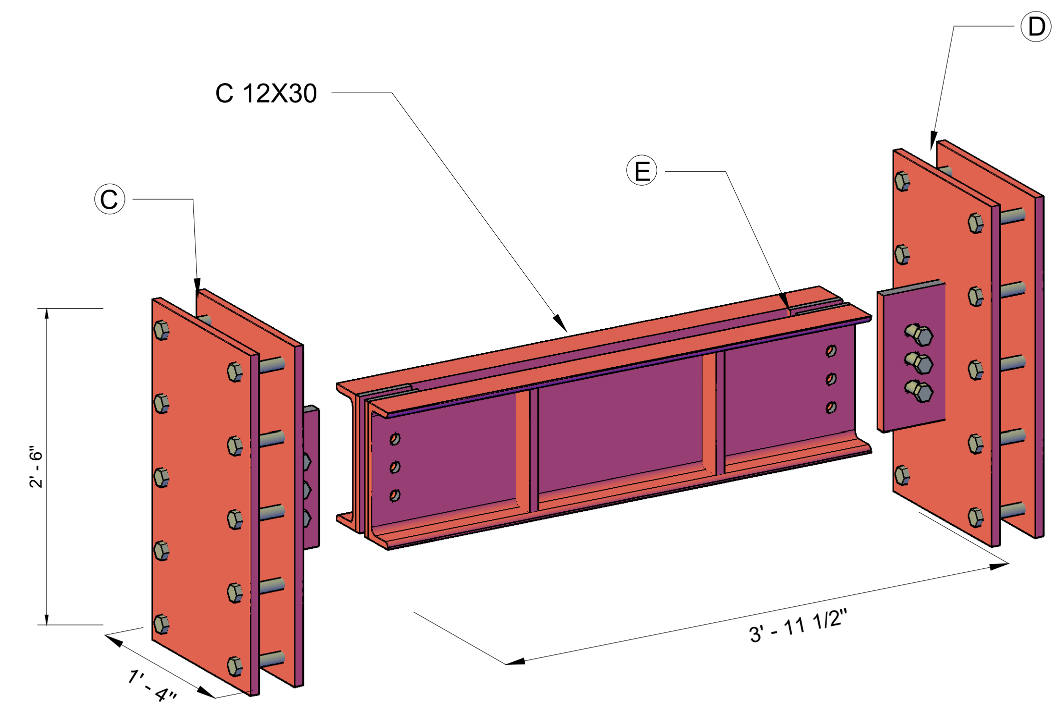
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

PT Anchor - Type I

SHEET NO.
S8-1

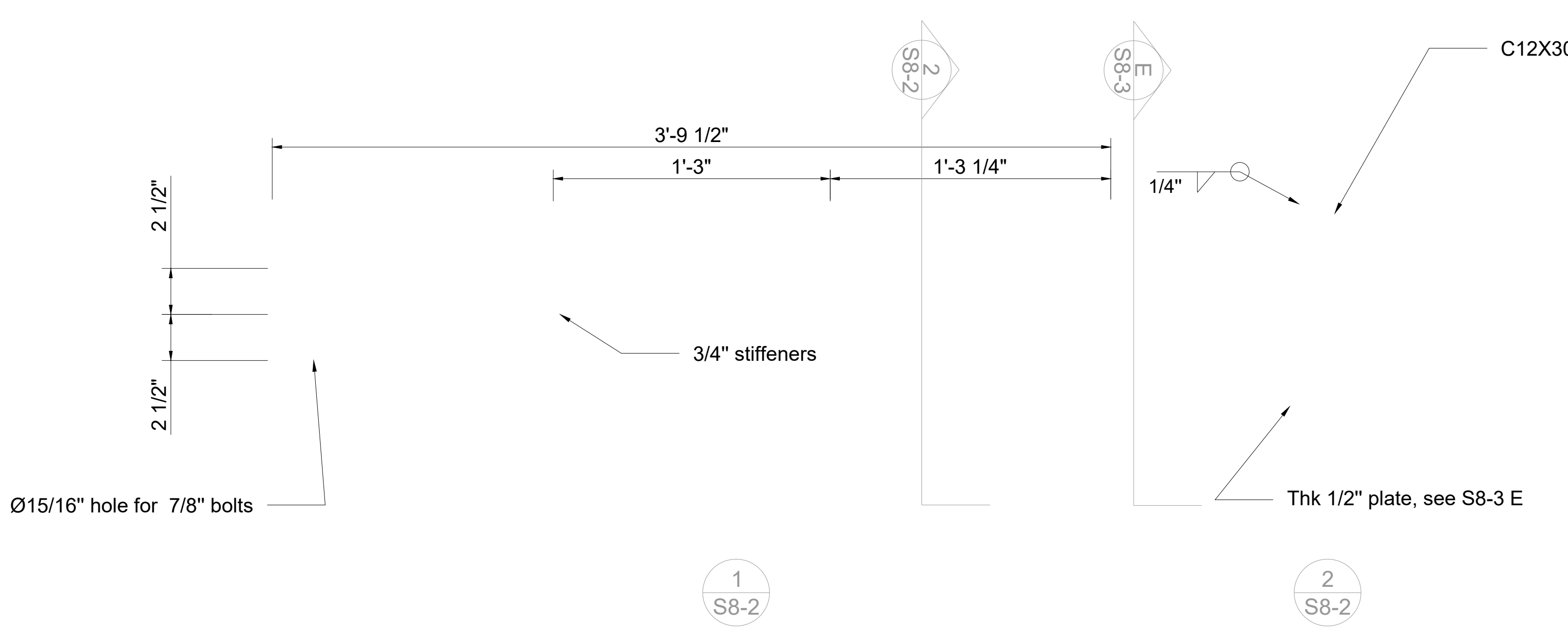
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Type II

Qyt: 1



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Smart Structures

ILEE

ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

PT Anchor - Type II

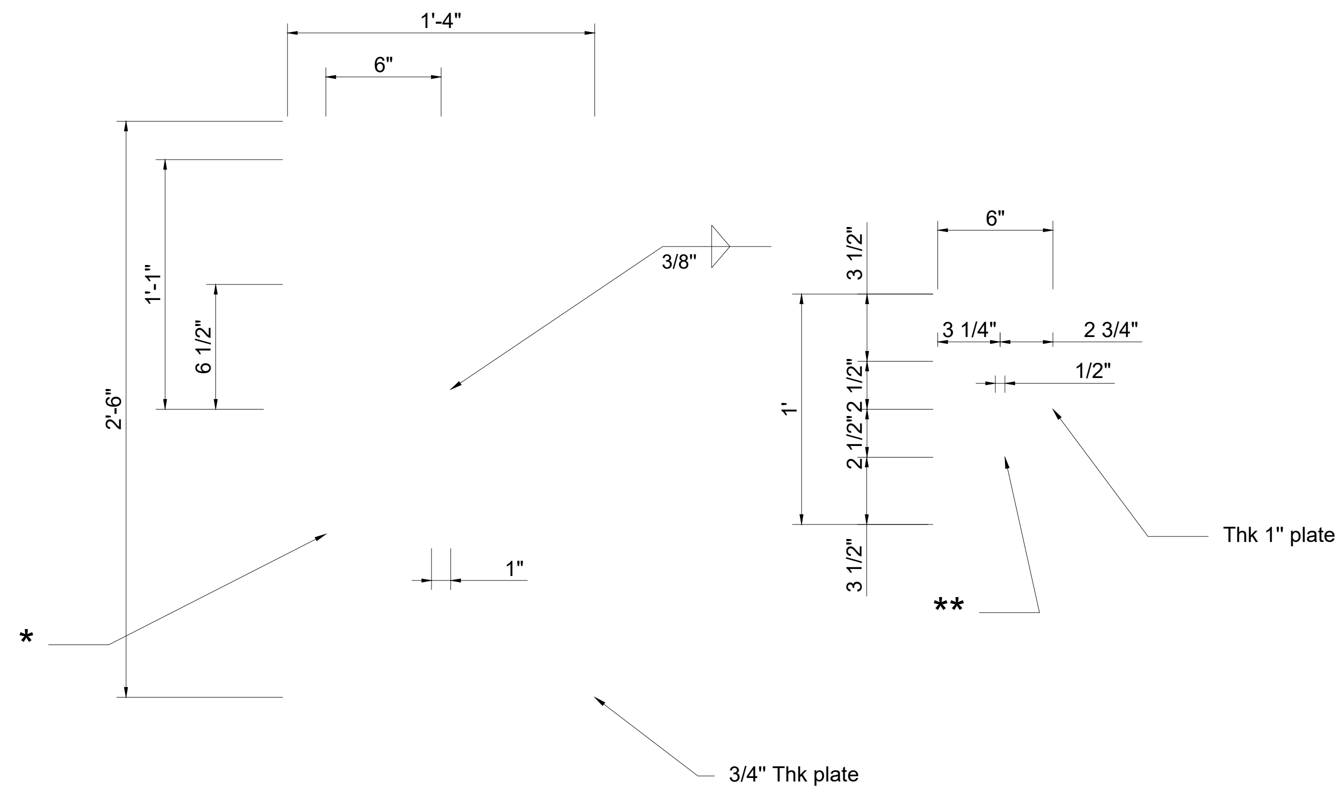
SHEET NO.

S8-2

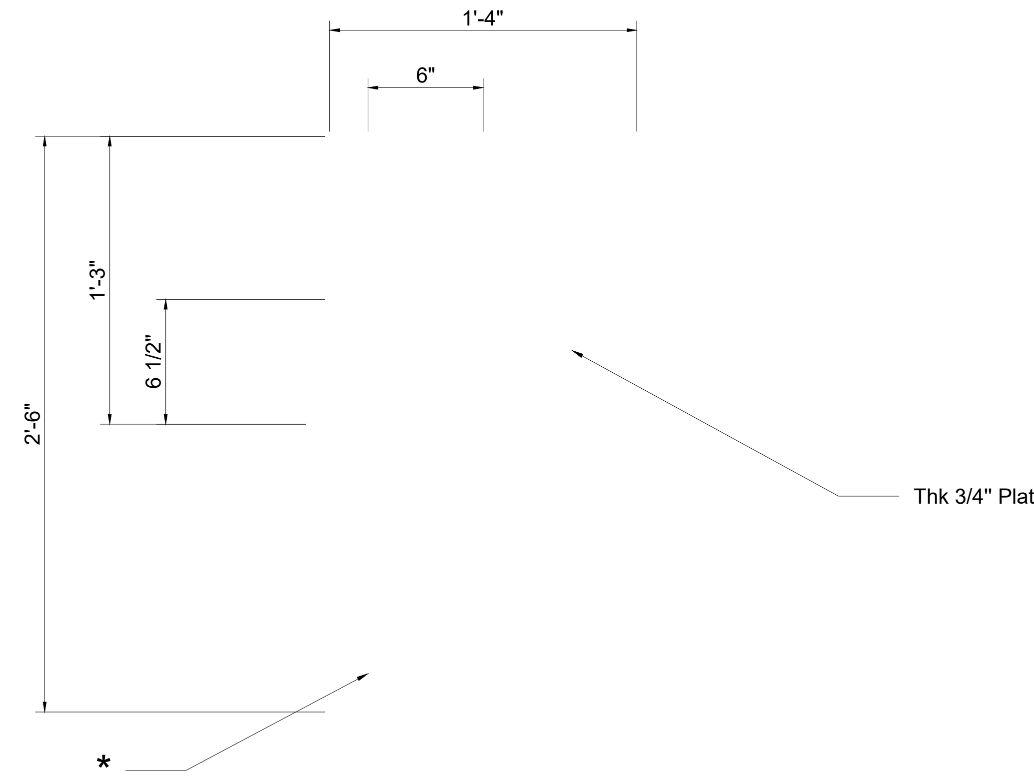
DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



C

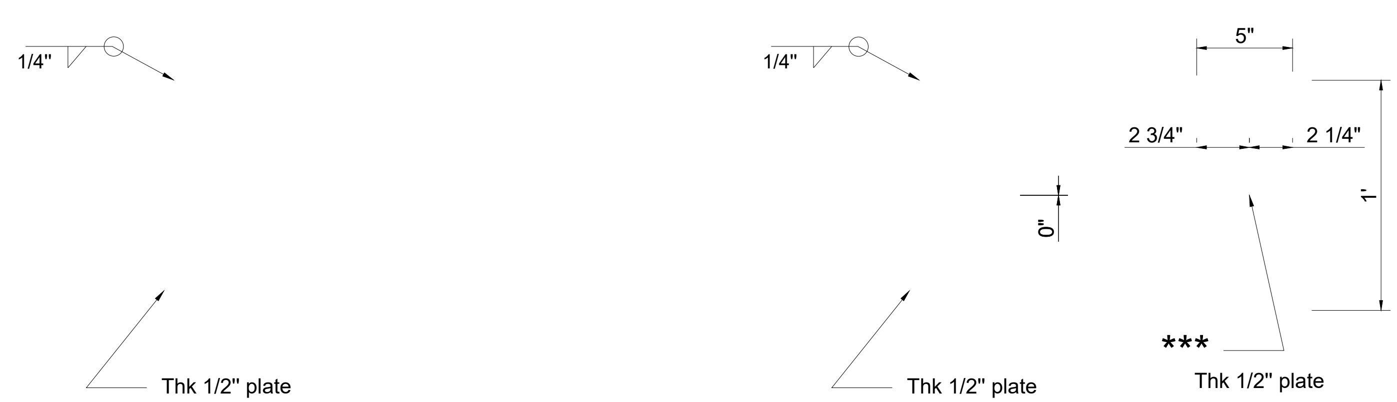


D

**Notes:*
If possible, use A325M bolts instead of rods

* Ø15/16" hole for 7/8" threaded rods

** Slot with Ø1/16" clearance for 7/8" bolt



E

*** Ø15/16" hole for 7/8" bolt

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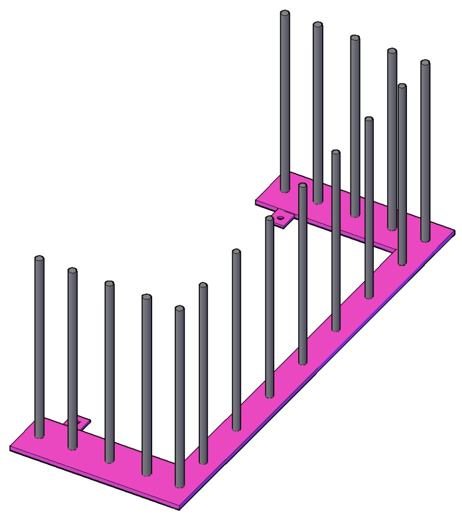
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

PT Anchor - Back Plate

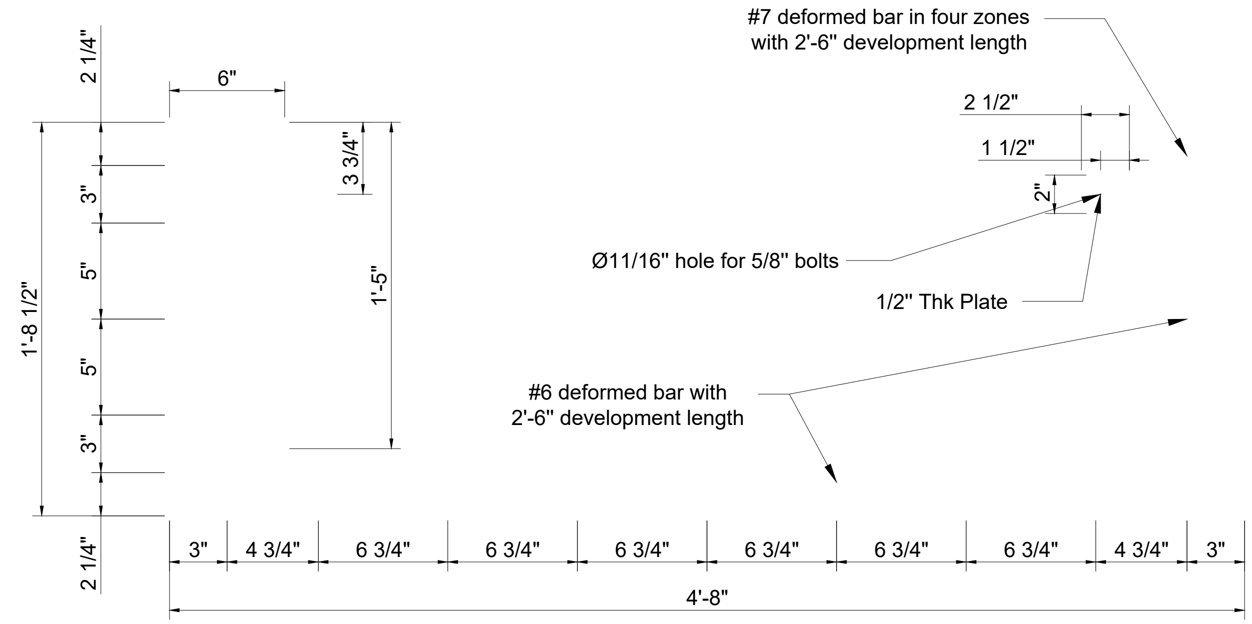
SHEET NO.
S8-3

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE	2023/10/08



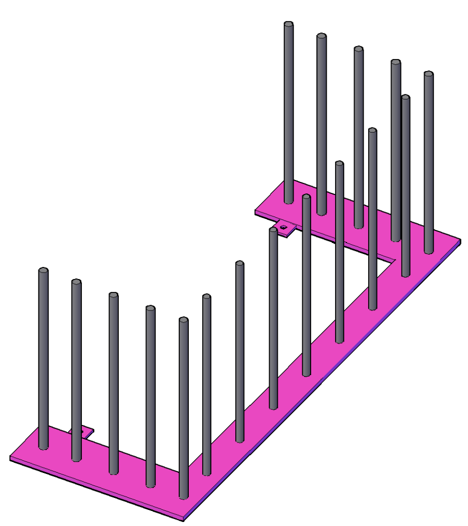
Wall to wall connection plate - Type I
3/4'' Thk plate
Qyt: 2
Unit Weight: 0.38kN/0.083kips



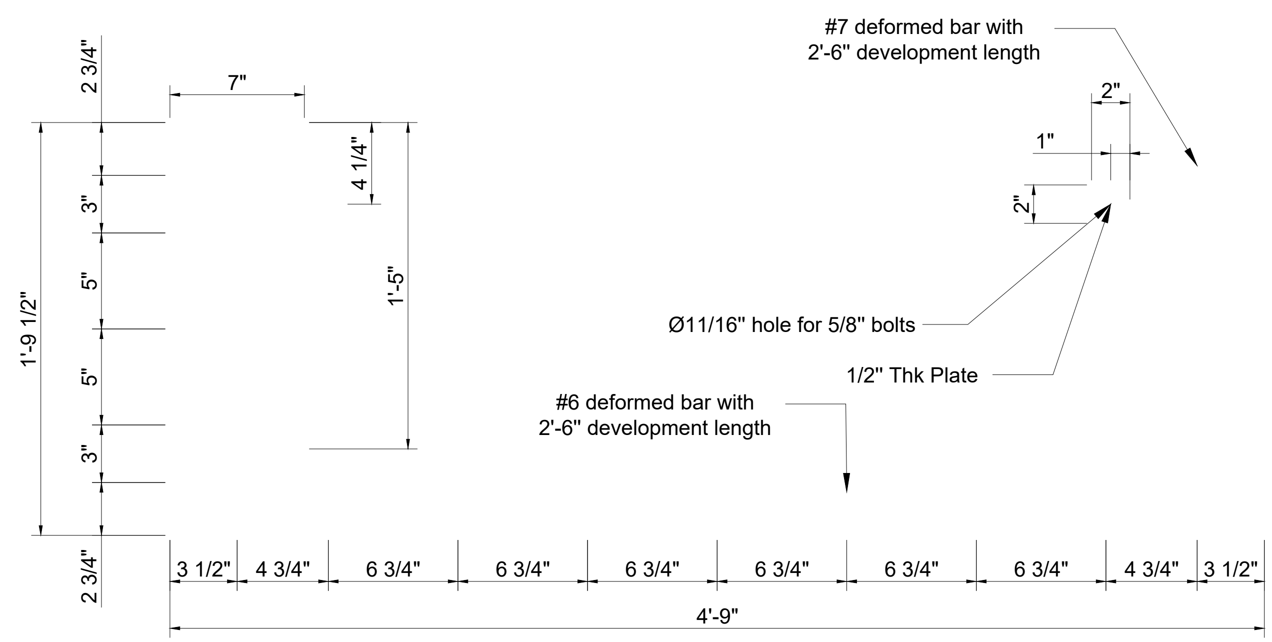
Note: Weld to develop fulls strength of rebars

1/S9 Thk 3/4" Plate

Qty. 1



Wall to wall connection plate - Type II
3/4'' Thk plate
Qyt: 2
Unit Weight: 0.44kN/0.098kips



Note: Weld to develop fulls strength of rebars

2/S9 Thk 3/4" Plate

Qty. 1

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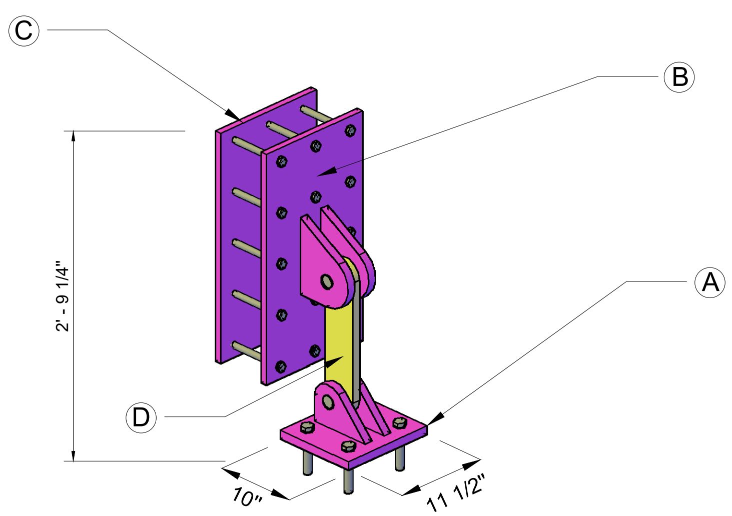
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Wall to Wall Connection

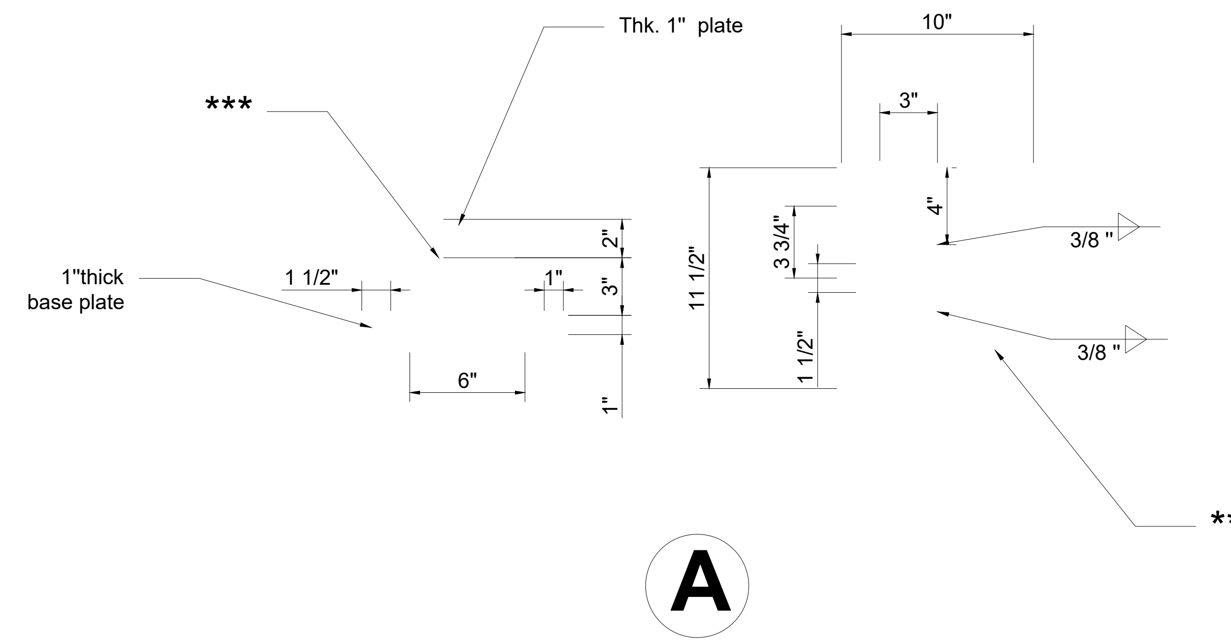
SHEET NO.
S9

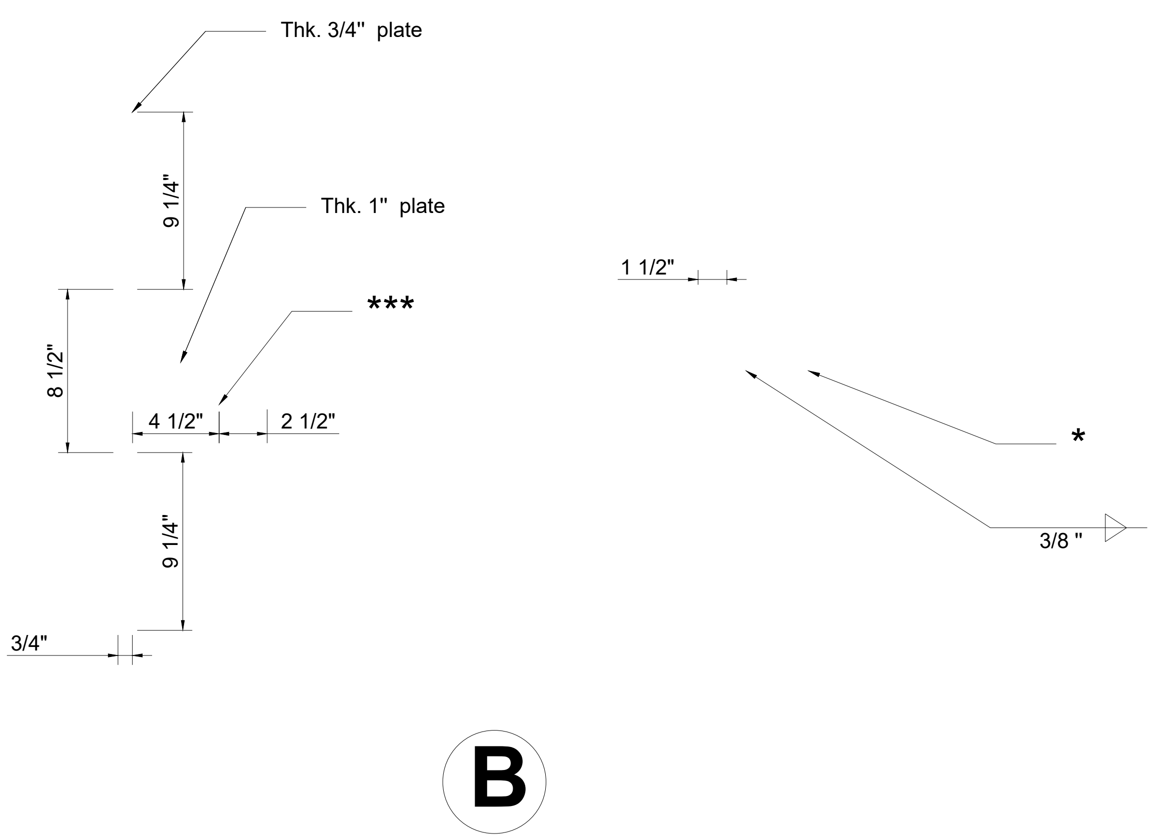
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

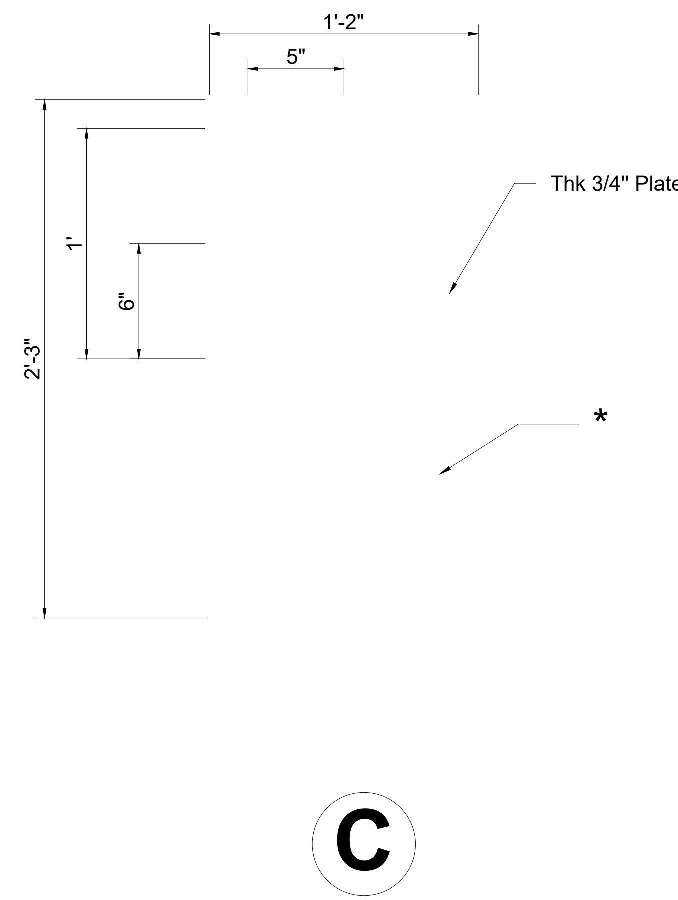
DATE 2023/10/08

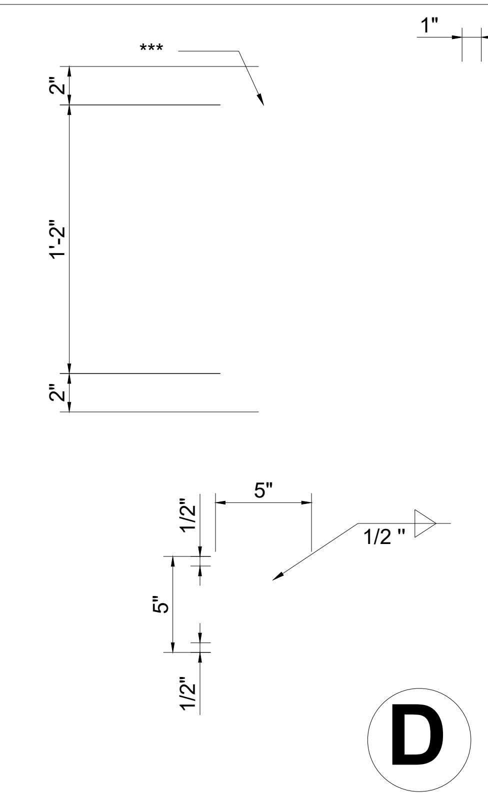


Qyt: 4









/ Notes:*
If possible, use A325M bolts instead of rods

* Ø11/16" hole for 5/8" threaded rods

** Ø1 1/16" holes for 1" all-threaded rods

*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

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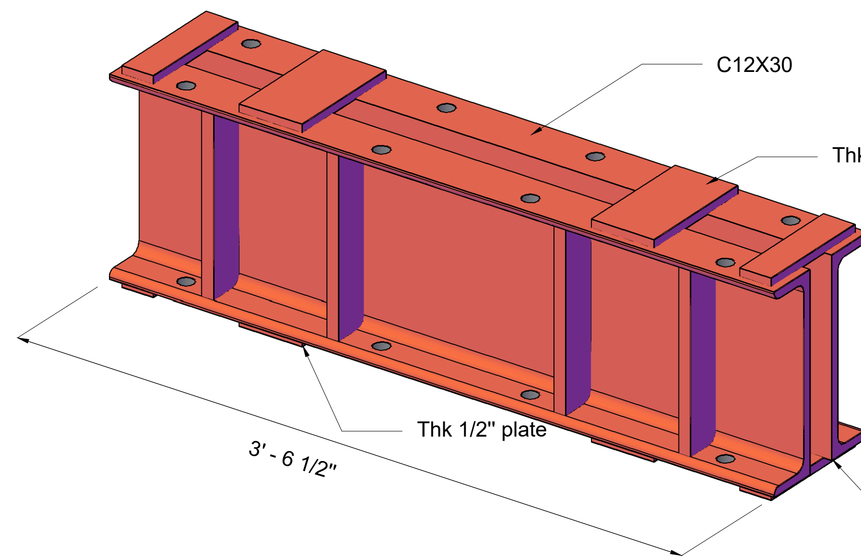
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

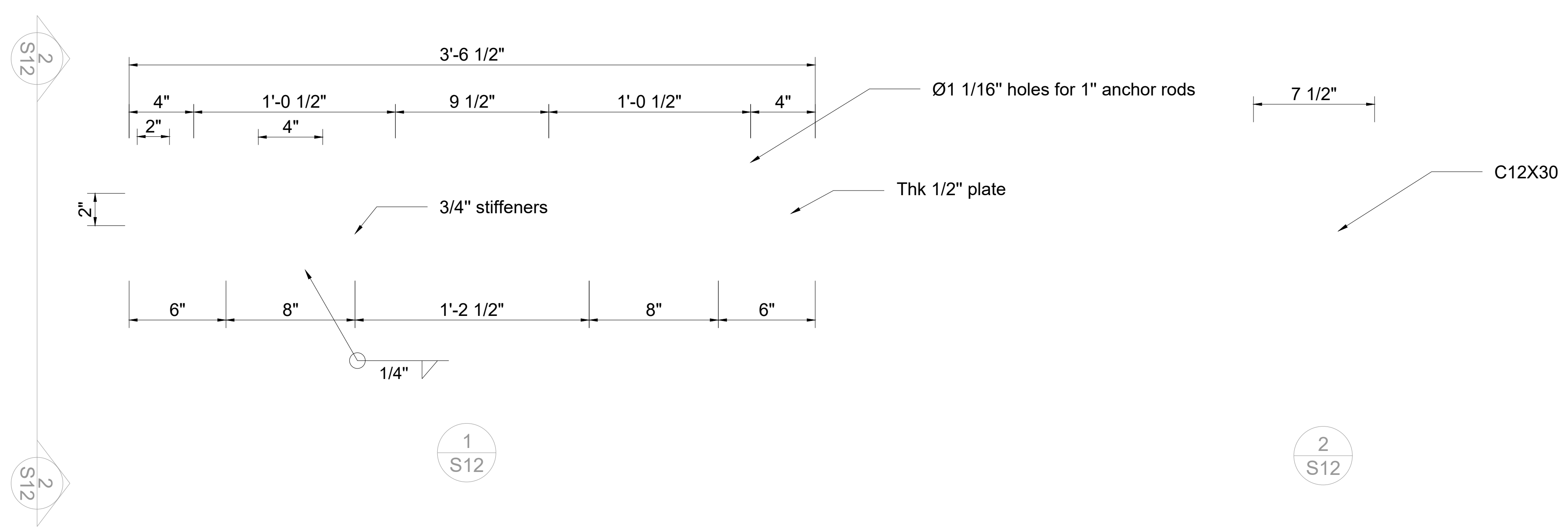
SHEET NO.
S10

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Qyt: 1



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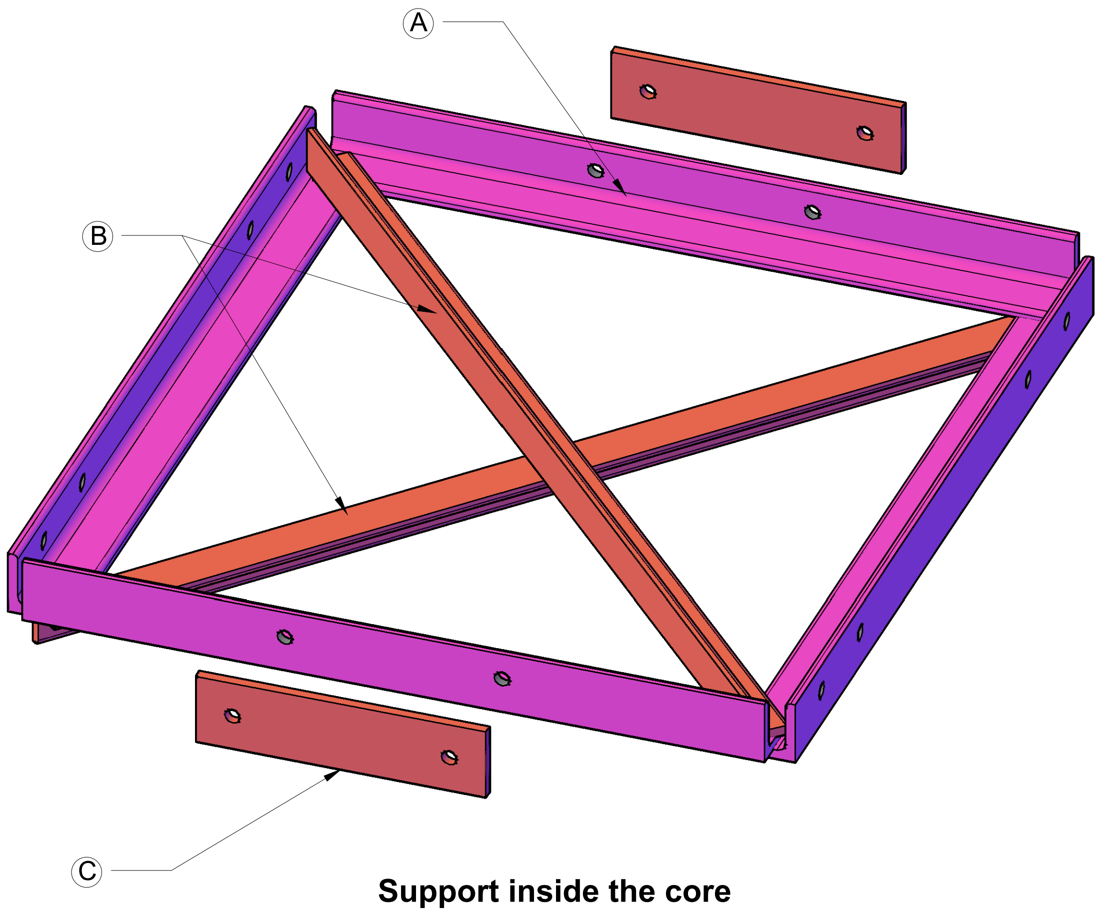
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

PT Base Anchor Plate

SHEET NO.
S11

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

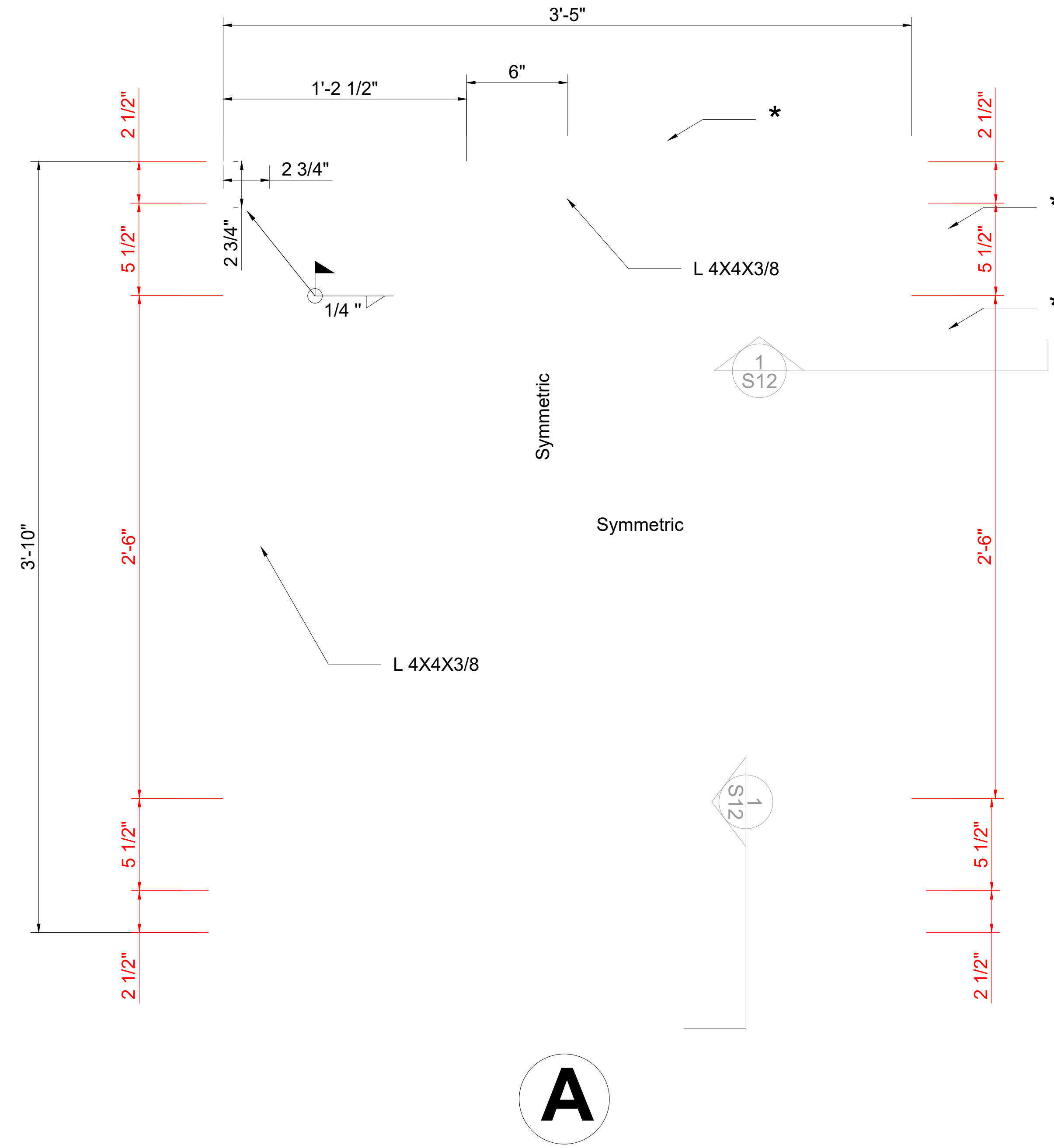
DATE 2023/10/08

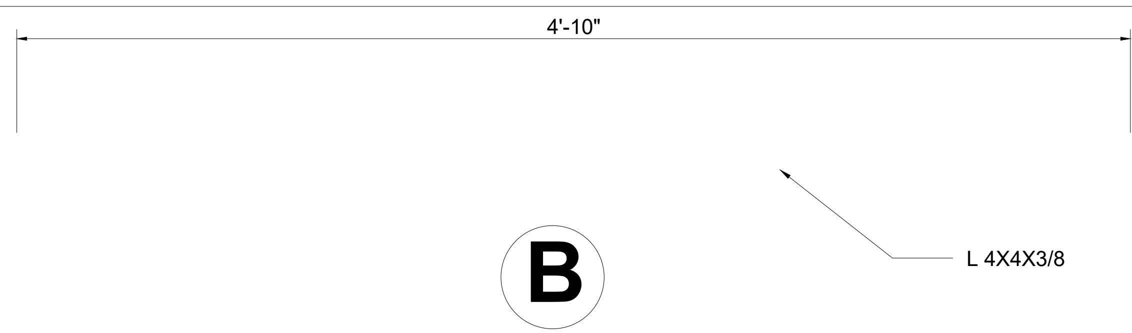


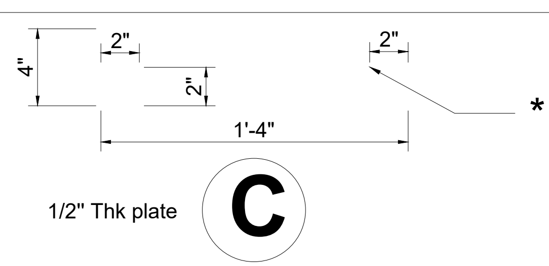
Support inside the core

Qyt: 8

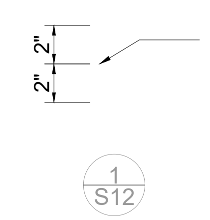
Notes: Hold welding until completion of concrete core







1/2" Thk plate



* Ø1 1/8" hole for 7/8" bolts

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ILEE

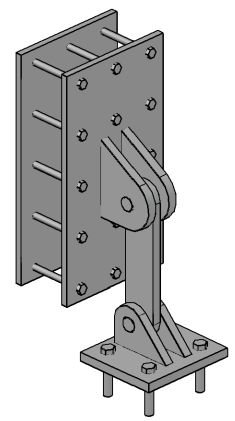
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Support inside the core

SHEET NO.
S12

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/30



UNDER REVISION

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Temporary Truss for Lifting

SHEET NO.

S12

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08

8 #3

#2 @ 1 3/4 " V.E.F
#2 @ 3 1/2 " H.E.F

#2 @ 4 1/2 " V.E.F
#2 @ 4 1/2 " H.E.F

8 #3

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Wall to Wall Connection

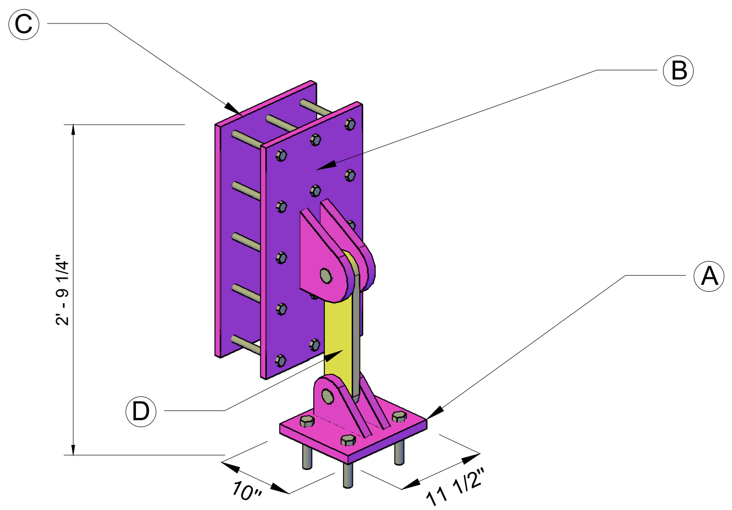
SHEET NO.

S9

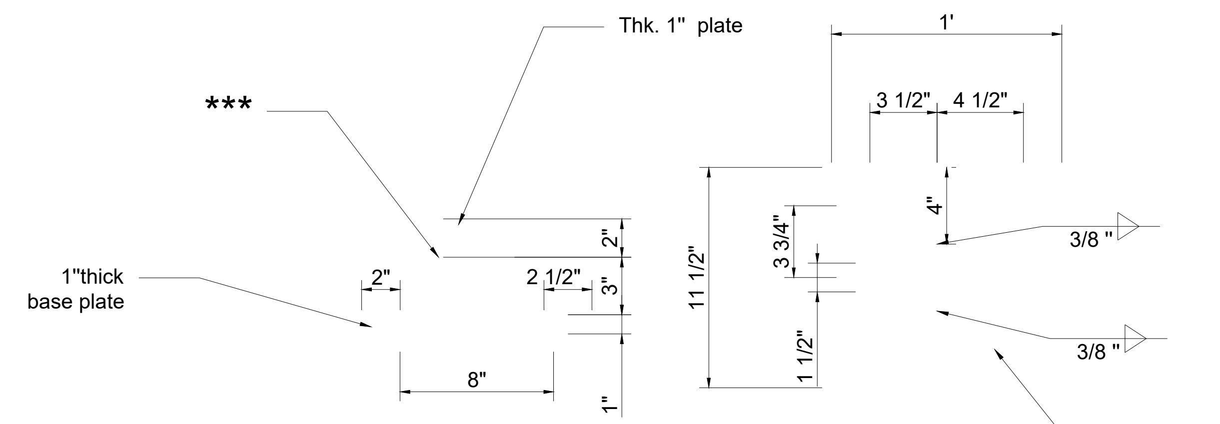
DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Qyt: 4



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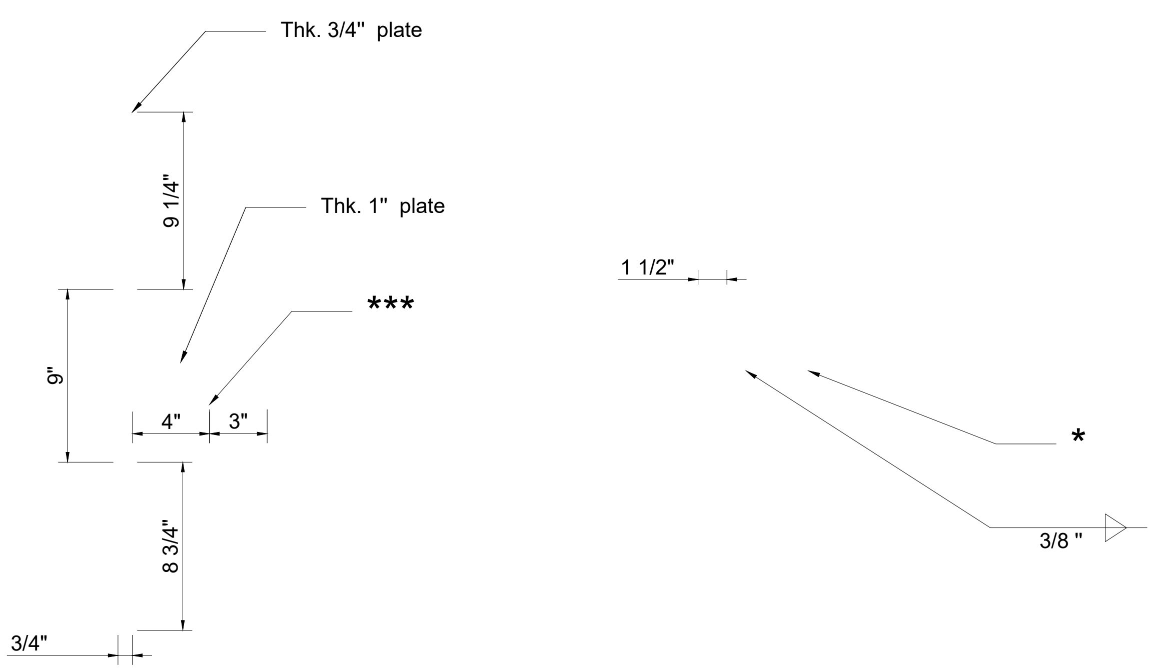

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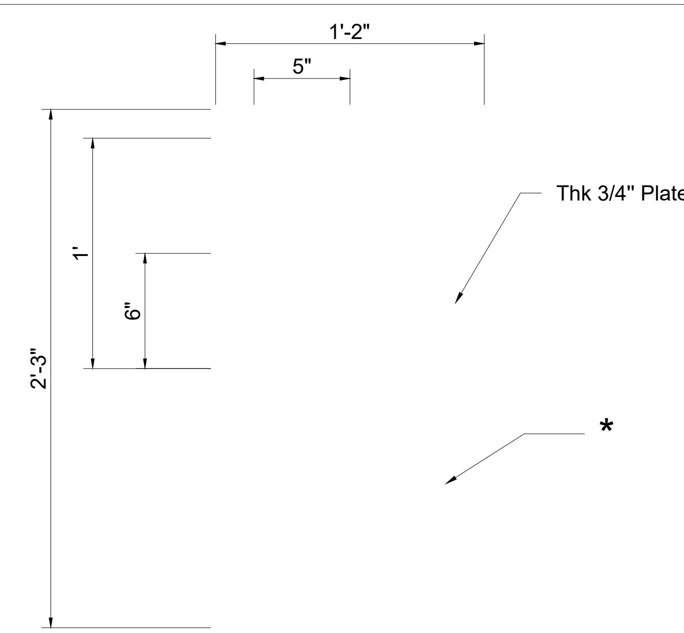
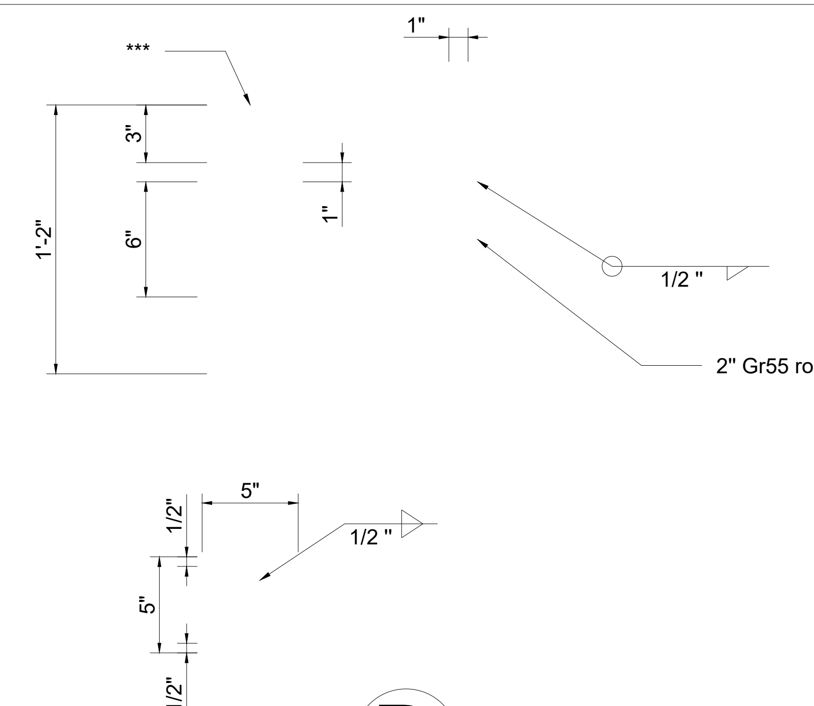


C



D

*/** Notes:
If possible, use A325M bolts instead of rods

* Ø11/16" hole for 5/8" threaded rods

** Ø1 1/16" holes for 1" all-threaded rods

*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

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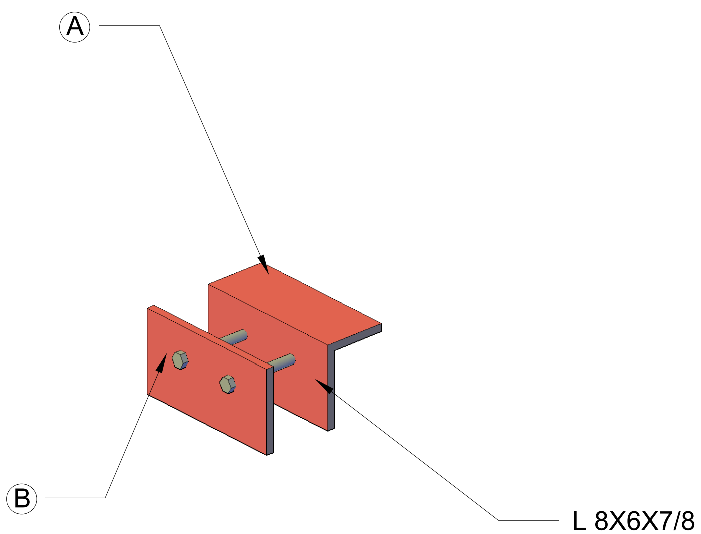
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

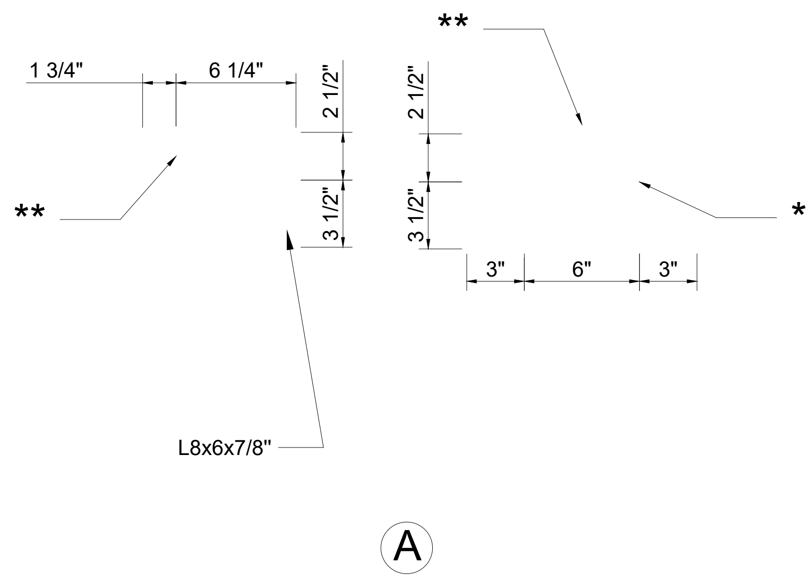
SHEET NO.
S10-R1

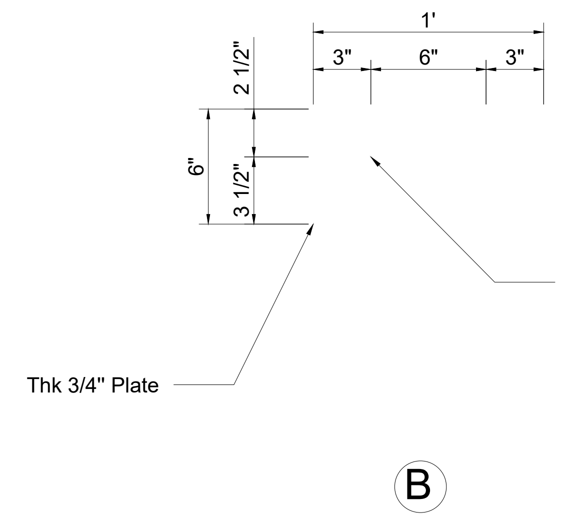
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/07



Qyt: 32





* Ø15/16" hole for 7/8" bolts

** Ø1 1/16" hole for 1" bolts

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

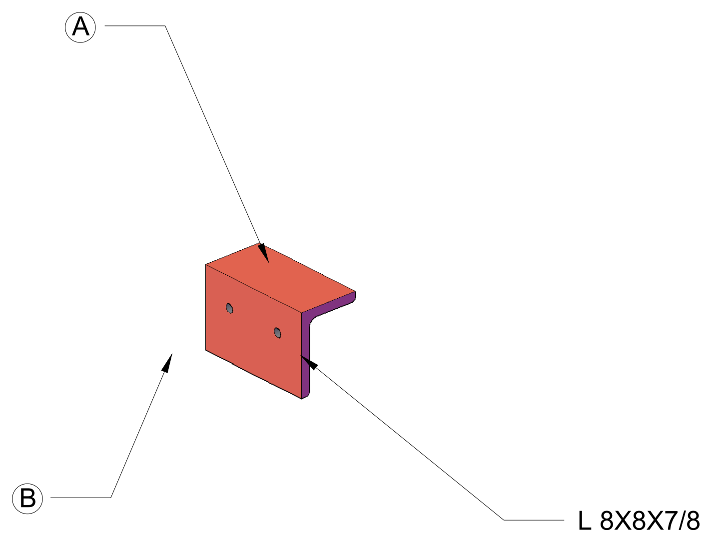
SHEET NO.

S7-R1

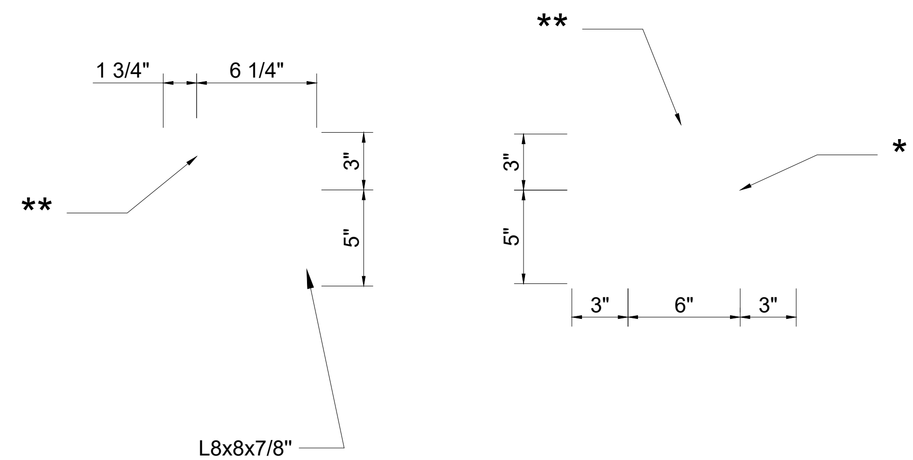
DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE	2018/11/08



Qyt: 32



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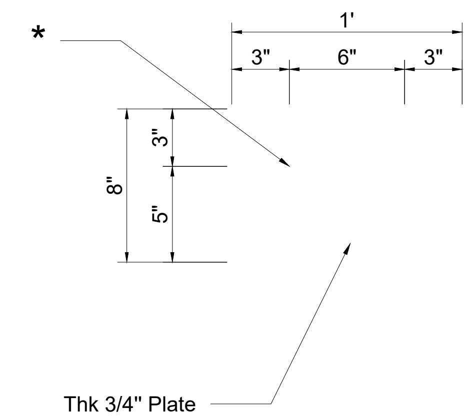

B

* Ø15/16" hole for 7/8" bolts

** Ø1 1/16" hole for 1" bolts

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

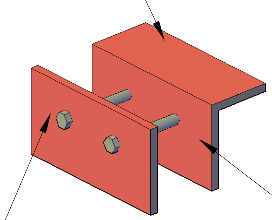
SHEET NO.

S7-R2

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/08



Qyt: 32

*

L8x8x7/8"

A

*

Thk 3/4" Plate

B

* Ø15/16" hole for 7/8" bolts

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

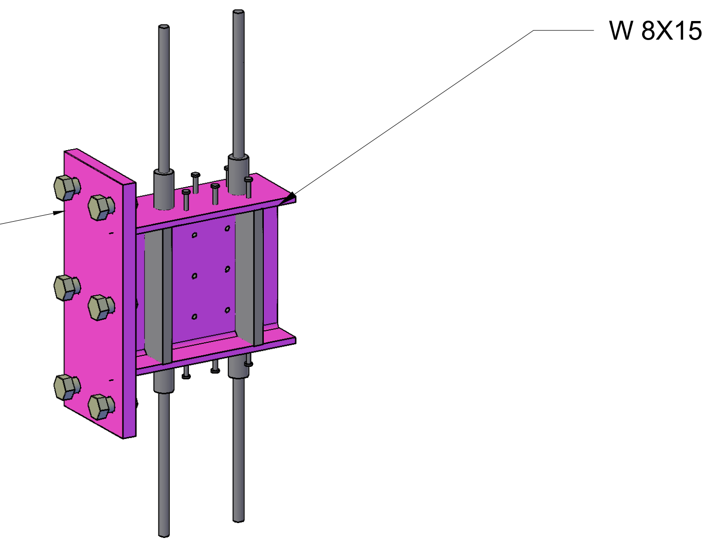
Mass Block Support & Rebar End Zone Plate

SHEET NO.

S7-R3

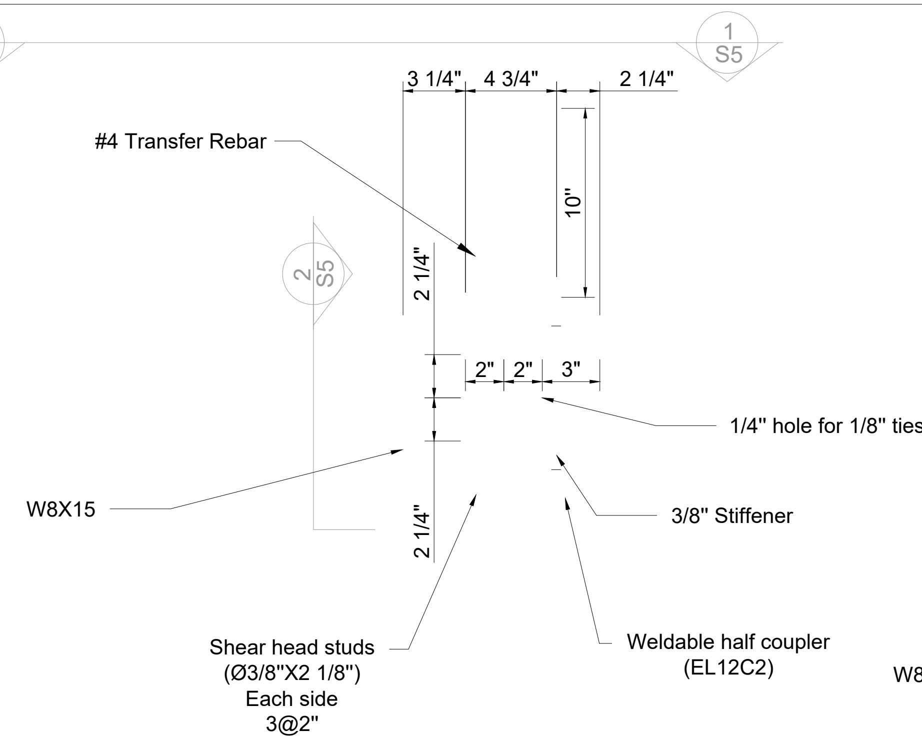
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

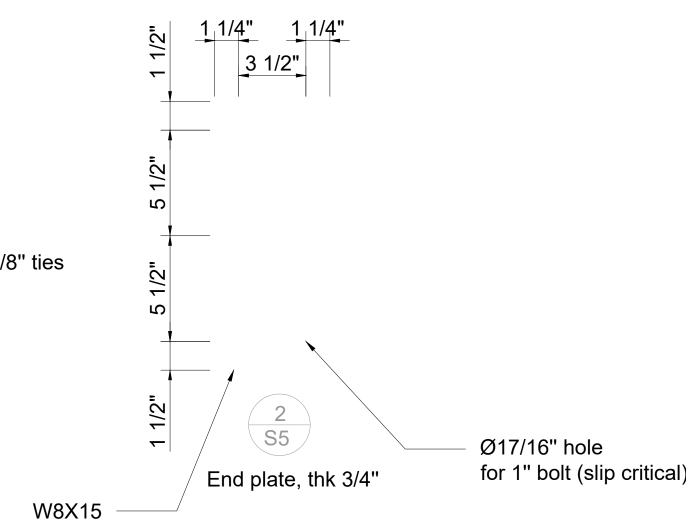
DATE 2018/11/08

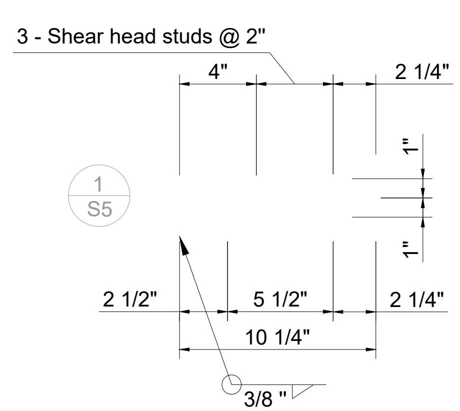


Notes for GTS:

Based on the conversation with Rob, we need steel components to connect the beam to the form work. Please add as you see fit.







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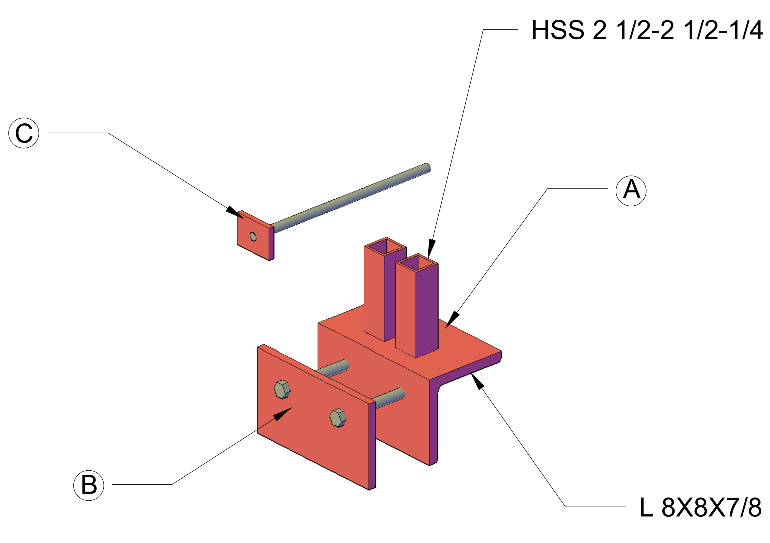
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Steel Embedded Beam

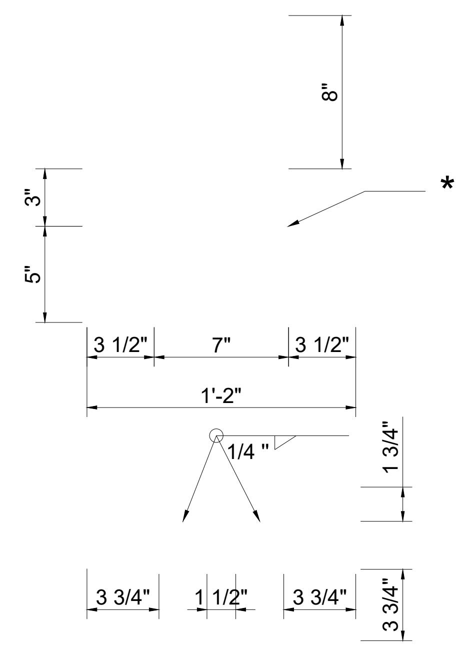
SHEET NO.
S5-R1

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE	2018/11/10



Qyt: 32



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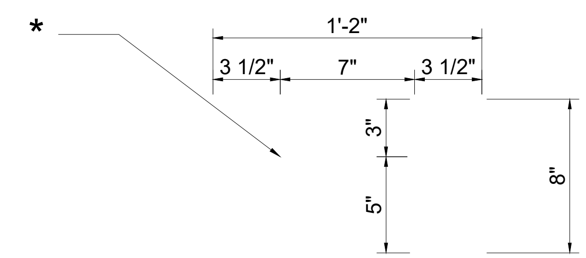
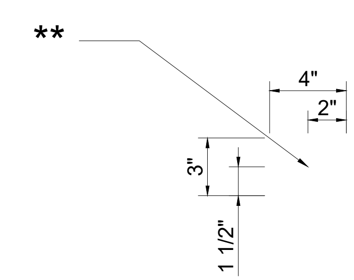


Thk 3/4" Plate

B



Thk 1/2" Plate, welded to the rebar, cast in mass block

C

* Ø15/16" hole for 7/8" bolts

** #6 rebar welded on, drill a hole if necessary

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST

NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

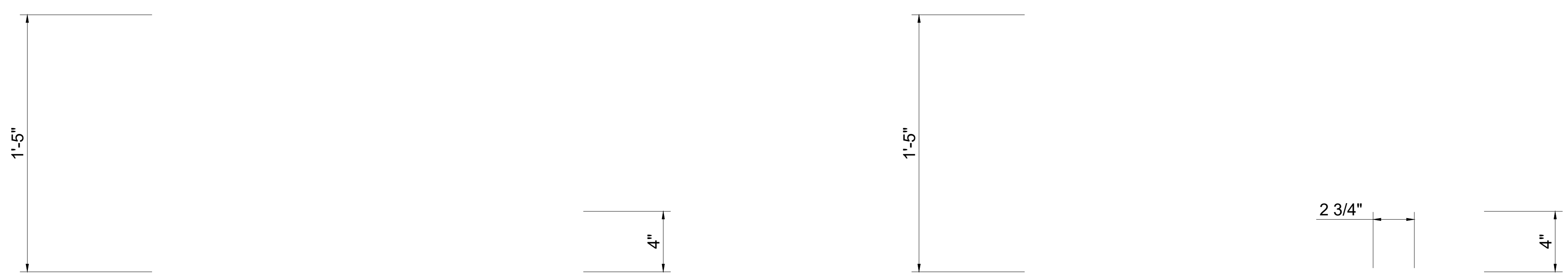
SHEET NO.

S7-R3

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/11



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NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

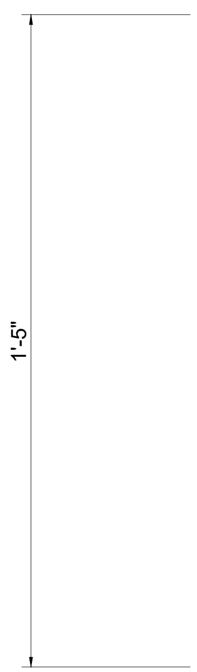
SHEET NO.

S7

DRAWN BY

Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/11



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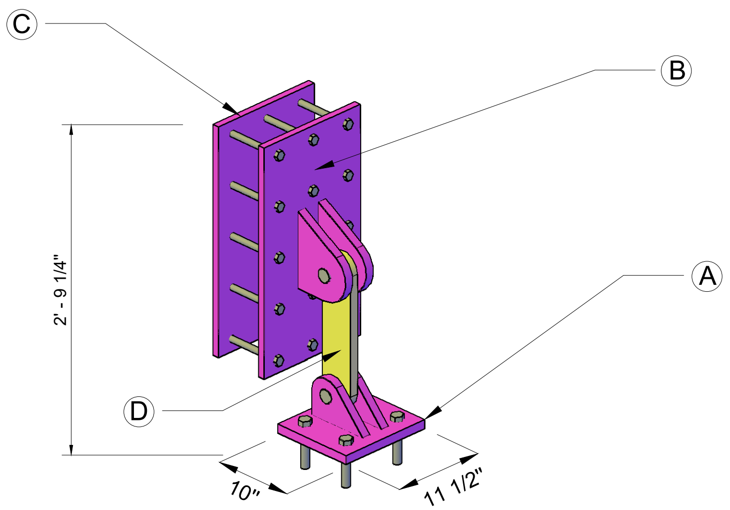
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support & Rebar End Zone Plate

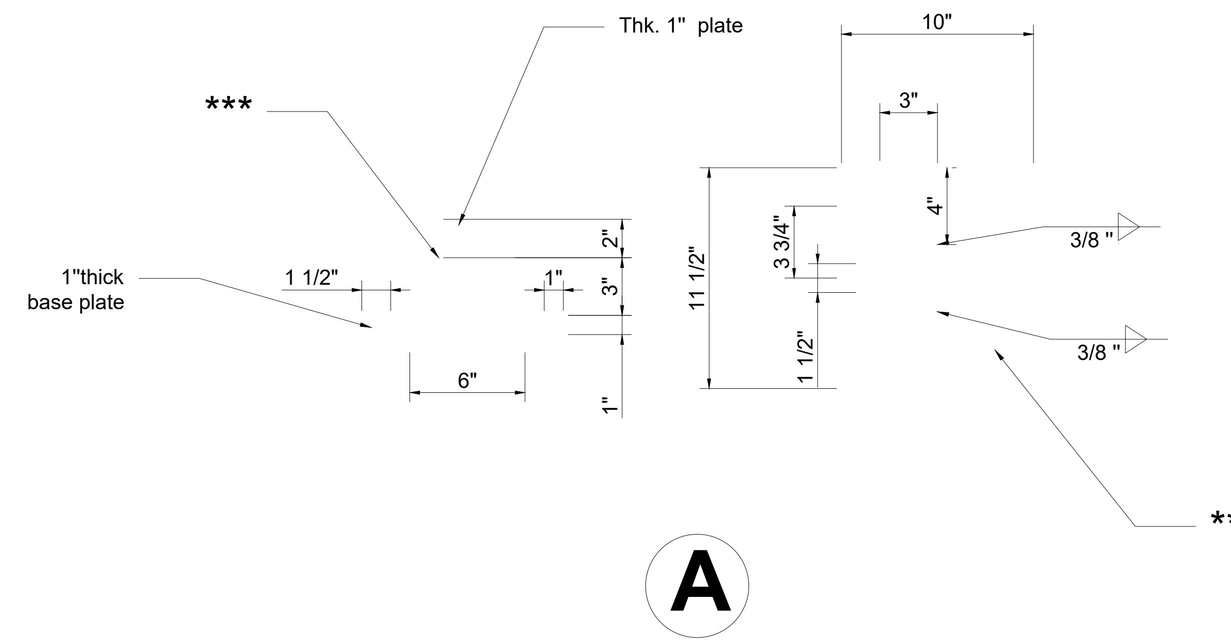
SHEET NO.
S7

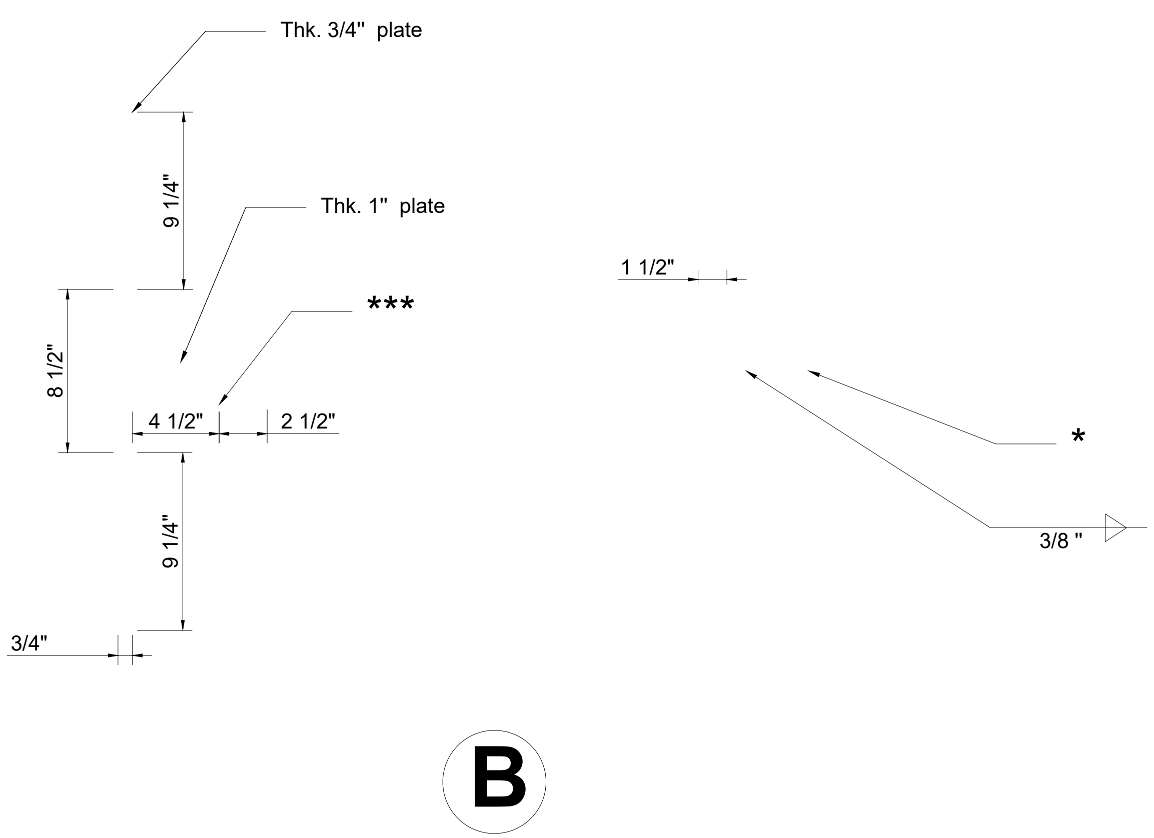
DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

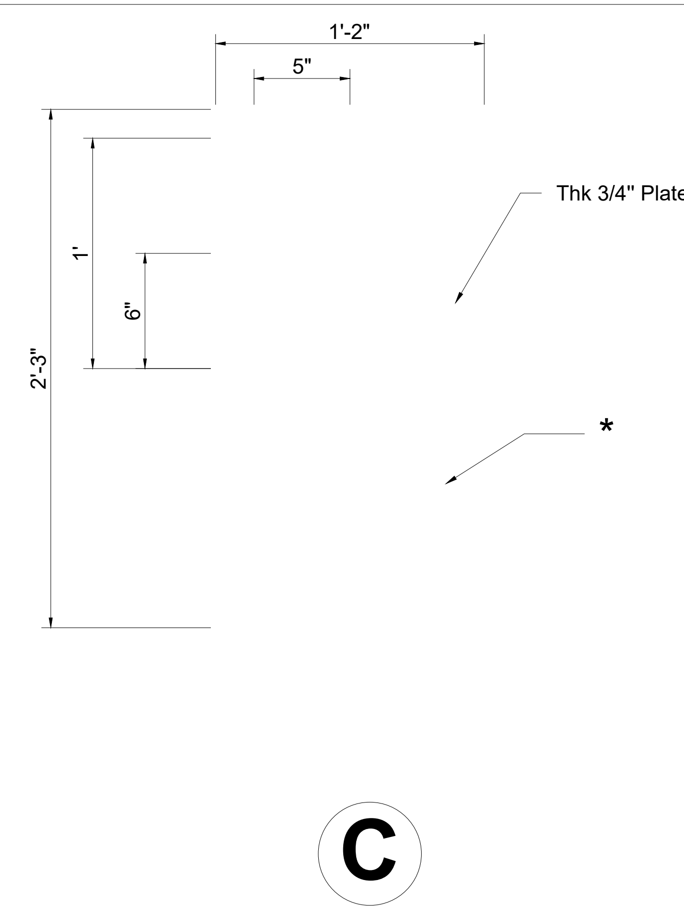
DATE 2018/11/11

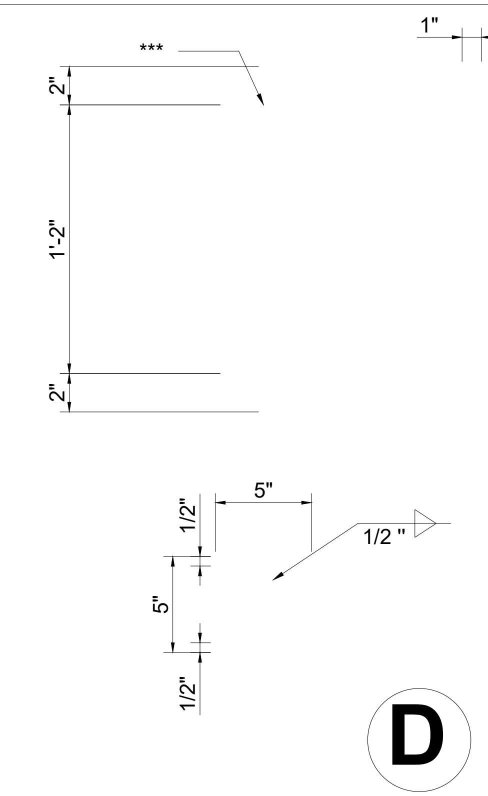


Qyt: 4









/ Notes:*
If possible, use A325M bolts instead of rods

* Ø11/16" hole for 5/8" threaded rods

** Ø1 1/16" holes for 1" all-threaded rods

*** Drill Hole for Ø 1 1/2" pin with tolerance less than 1/32", the less the better

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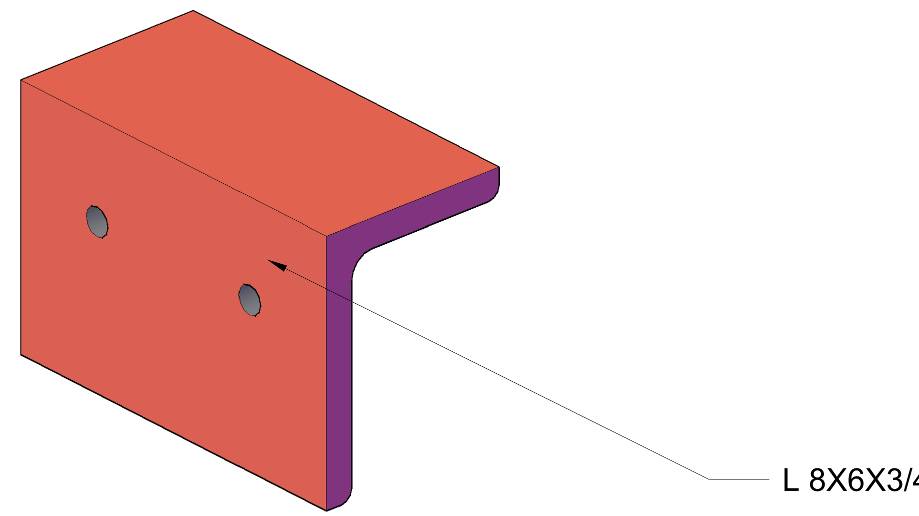
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

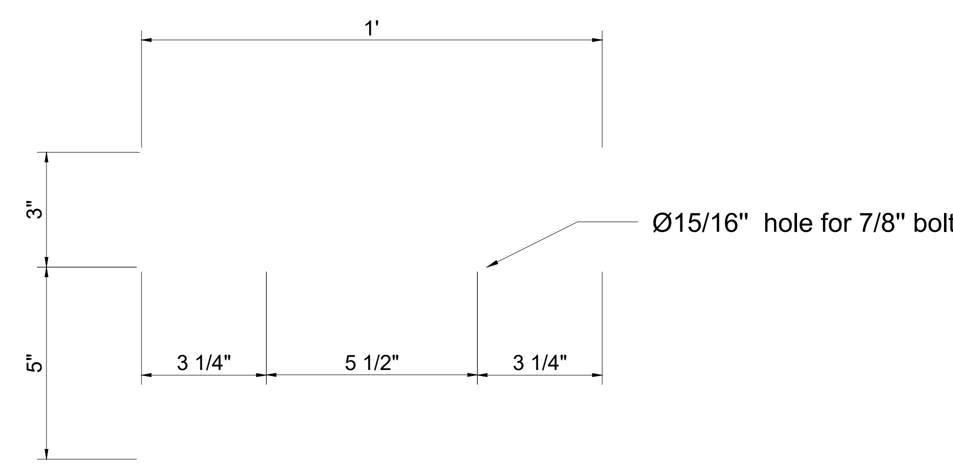
SHEET NO.
S10-R1

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2023/10/08



Qyt: 32



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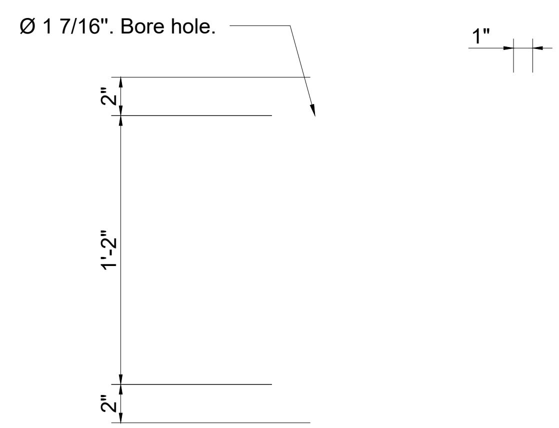
ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Mass Block Support

SHEET NO.	**M109R1**
DRAWN BY	Tianyang Qiao Hengchao Xu Lisa Tobber
DATE	2018/11/30



Qyt: 8

M121

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ILEE-EERF COLLABORATIVE SHAKE-TABLE TEST
NEXT-GENERATION EARTHQUAKE RESILIENT TALL BUILDING

Base Fuse Connection

SHEET NO.
M121R1

DRAWN BY
Tianyang Qiao
Hengchao Xu
Lisa Tobber

DATE 2018/11/30